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Open RAN for 6G Networks: Architecture, Use Cases and Open Issues

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***Abstract*—Enhancing the radio access network (RAN) architecture has been the focus of some of the latest global operators' concentrated effort, building on principles of intelligence and openness. The goal is to develop an operator-defined RAN architecture (as well as related interfaces) on an open hardware that will enable intelligent radio control for the Fifth Generation (5G) networks beyond, especially the Sixth Generation (6G) wireless networks. The transition to 6G is anticipated to be made simpler by this Open RAN architecture. The reduced cell radius of the 6G mobile networks, which will use significantly higher frequencies, will severely limit their ability to penetrate buildings. Radios will need to be positioned closer to subscriber devices and indoor installations will need to be used in order to accommodate these changes. Operators will require a greater variety of form factors and potentially even new types of front-haul in order to deploy affordably large numbers of Radio Units (RU) indoors. By having open interfaces between the Distributed Unit (DU) and RU, Mobile Network Operators (MNOs) will have access to a much wider range of RUs with various form factors and customizable features. This survey summarizes the different possible architectures for 6G Open RAN, especially considered from the perspective of energy efficiency, and low latency. Next, it presents various novel use cases which can employ 6G Open RAN architectures. Lastly, the prospects and problems facing Open RAN in 6G are discussed. Design principles and potential fixes for sophisticated processes are also provided.**

***Index Terms*—6G Network, AI/ML Algorithms, Open RAN Architecture, Low Latency, Energy Efficiency, Digital Twinning**

NOMENCLATURE

3GPP	3rd Generation Partnership Project
6G	Sixth Generation
AD	Anomaly Detection
AI	Artificial Intelligence
AR	Augmented Reality
ASIC	Application Specific Integrated Circuit
BBU	Base Band Unit
BS	Base Station
BWP	Bandwidth Part
C-RAN	Centralized Radio Access Network
CI/CD	Continuous Integration and Continuous Deployment
CN	Core Network
CNF	Cloud Native Function
COTS	Commercialized-off-the-shelf
CP	Control Plane
CPU	Central Processing Unit
CQI	Channel Quality Index
CU	Centralized Unit
D-RAN	Distributed Radio Access Network
DPU	Data Processing Unit
DQN	Deep Q-Network
DT	Digital Twinning
DU	Distributed Unit
EE	Energy Efficiency
eMBB	enhanced Mobile Broadband
FCAPS	Fault, Configuration, Accounting, Performance, Security
FEC	Forward Error Correction
FH	FrontHaul
FL	Federated Learning
FPGA	Field Programmable Gate Array
GPP	General Purpose Processor
HLS	High Layer Split
HW	Hardware
HWA	Hardware Accelerators
IAB	Internet Access Backhaul
IoT	Internet of Things
KPI	Key Performance Indicators
LDPC	Low Density Parity Check
LL	Low Latency
LLS	Lower Layer Split
LTE	Long Term Evolution
MEC	Mobile Edge Computing
ML	Machine Learning
mMIMO	massive Multiple Input Multiple Output
MNO	Mobile Network Operator
NBI	North Bound Interface
Near-RT RIC	near-Real-time RAN Intelligent Controller
NF	Network Function
NN	Nomadic Nodes
Non-RT RIC	Non-Real-time RAN Intelligent Controller
O-RAN	Open RAN
PNF	Physical Network Function
QC	Quantum Computing
QoE	Quality of Experience
QoS	Quality of Service
RAN	Radio Access Network
REM	Radio Environment Map
RF	Radio Frequency
RIC	RAN Intelligent Controller
RL	Reinforcement Learning
RRM	Radio Resource Management
RSRP	Received Signal Reference Power
RU	Radio Unit
SDR	Software Defined Radio
SINR	Signal-to-Noise-Ratio
SMO	Service Management Orchestration
SNC	Stochastic Network Calculus
SoC	System on Chip
SW	Software
UE	User Equipment
UL	Uplink
UP	User Plane
URLLC	Ultra Reliable Low Latency Communication
v-RAN	Virtualized Radio Access Network
V2I	Vehicle-2-Infrastructure
V2N	Vehicle-2-Network
V2P	Vehicle-2-Pedestrian
V2V	Vehicle-2-Vehicle

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V2X Vehicle to Everything
VNF Virtual Native Function
VR/XR Virtual/Extended Reality

I. Introduction

TEN years ago someone said that "if we get 5G right, there will be no need for 6G" [1]. Yet, despite all the efforts put into 5G worldwide, which turned 5G into a great commercial success, 6G is already being designed by the scientific community. There is also a growing expectation that the newer, faster, better 6G will eventually replace 5G, sooner or later.

A. 6G in the Context of the NextG Network

In the context of a *NextG* network to be released, regardless of its actual number, the following points are worth noting.

1) In the **past**, every odd generation introduced a "new" type of service to mobile communications (i.e. 1G supported voice services in mobile communications, 3G proposed the Internet access, and 5G introduced native incorporation of verticals), while every even generation offered a better solution for the odd generation services (i.e. 2G improved voice communications and 4G offered better Internet access). That is why, roughly speaking, 6G may be required to correct the 5G shortcomings.
2) **Technically speaking**, an *UltimateG* might exist; nevertheless, this does not imply that it is the final answer to every communication-related issue faced by humanity. This indicates that, as software-based solutions become prevalent, the *UltimateG* can be easily expanded to accommodate any new technology that emerges in the future. The challenge is ensuring that the *UltimateG* design is performed correctly, which is very challenging, given the anticipation, excitement and deadlines imposed in the process of developing each *NextG*.
3) There is also a strong **marketing component** to consider, as vendors must regularly offer the *NextG* to the operators and the operators look forward to offering their clients a *NextG* service. Therefore, it seems that there will always be a need for a *NextG*.

In this context, compared to services over the current 5G networks, the new 6G applications will have many additional features and therefore they will require larger and more diverse network support. Increased network capacity, reduced latency and increased reliability are some fundamental 6G network requirements. More precisely, as wireless communications are one of the critical components of our life, both in relation to the development of economic sectors and social interaction, an important focus should be on advancements of next-generation wireless networks. Consequently, both the scientific community and industry are expected to put significant efforts and enhance these networks to achieve common goals [2]. Future communications should have the following key characteristics: system dependability, device sustainability, automation and digitalization for a simpler life, and infinite connectivity to meet application demands. It is anticipated that the next period will undergo significant technological advancements and 6G will serve as a crucial communication and information backbone. Anything should communicate with anything else, anywhere and anytime, in a 6G environment.

However, due to the requirements of modern applications, the future generation of wireless networks should accomplish several goals in order to support these applications in terms of their high QoS levels. Applications such as those involving virtual, augmented, and mixed reality, digital twinning (DT) or remote control of delicate operations, should be supported by the next-generation wireless networks in terms of extremely high data rates, very low end-to-end delays, extremely high reliability, massive cell capacity, and extended coverage [3]–[6]. It is suggested that 6G should target mostly the following key performance metrics:

- Very low latency communication (10 us-100us)
- Very high peak data rate (at least 1 Tb/s)
- Very high mobility ($\geq$ 1000km/h)
- Huge capacity (10^7 device/km^2)
- Reduced energy consumption to introduce green communications (100× reduction compared with 5G)
- Data rate up to 10Gb/s (10x higher than 5G).
- High spectrum efficiency (5–10x higher than 5G).
- High 3D coverage, including terrestrial, aerial, space, and sea.

In relation to the above mentioned KPIs, the analysts have pointed to the low commercial uptake of the more advanced 5G services that demand LL, and high peak rates. And the 3GPP submission for IMT-2020 (5G) recognised that data rates could, theoretically, be supported up to 140 Gbps. So it could be argued that 5G already supports the connectivity demanded by current customers [88].

B. Open RAN

A key element of any cellular network is the radio access network (RAN) [7]. The user equipment (UE) and the core network (CN) are connected by the RAN. Over time, the RAN component has changed to meet the needs of increasing user demands and growing number of subscribers. Currently, three types of RANs are widely used.

1) **Distributed Radio Access Network (D-RAN):** In a D-RAN, the base stations are geographically dispersed, and each base station contains its own processing and radio resources. This traditional architecture provides localized processing, but can be less efficient in terms of resource utilization and management.
2) **Centralized Radio Access Network (C-RAN):** A C-RAN centralizes the baseband processing units in a central location, while the radio units remain distributed at the cell sites. This centralization allows for better resource pooling, reduced operational costs, and improved performance through coordinated signal processing.
3) **Virtualized Radio Access Network (vRAN):** A vRAN takes the C-RAN concept further by virtualizing the network functions. Baseband processing is performed on virtual machines or containers, enabling more flexible, scalable, and cost-effective deployment and management. This approach supports dynamic allocation of resources and leverages the cloud infrastructure for enhanced efficiency.

Despite vRAN's important benefits such as flexibility, scalability, and cost efficiency through virtualization and cloud-based infrastructure, they face several challenges, mostly in terms of:

Performance: Virtualization can introduce additional latency and may not yet be able to match the real-time performance needs of 5G.

Complexity: Managing and orchestrating virtualized resources across a distributed network infrastructure is complex and can lead to inefficiencies.

Resource Demands: The intensive processing power and fast, reliable connectivity required for 5G services might exceed what current vRAN implementations can provide.

Maturity: vRAN technology is still evolving, and existing solutions might not be mature enough to handle the full spectrum of 5G and beyond use cases effectively.

In this context, there was a need to move forward and expand the RAN ecosystem, putting the basis of a more open, interoperable, and flexible RAN environment, opening the door for the Open RAN (O-RAN).

The O-RAN term was first coined by the O-RAN Alliance, a standardization organization with the goal of *"reshaping the RAN industry towards more intelligent, open, virtualized, and fully interoperable mobile networks"* [8]. Gathering a wide support beyond the companies involved in the original O-RAN ecosystem definition, the general notion of O-RAN went further, describing a complex concept. O-RAN aims to drive innovation, reduce costs, and improve the adaptability and performance of the network, preparing it to accept and overcome future challenges. Fig. 2 illustrates the foundations on which the O-RAN concept was built and of which different aspects will be discussed in detail in the subsequent sections.

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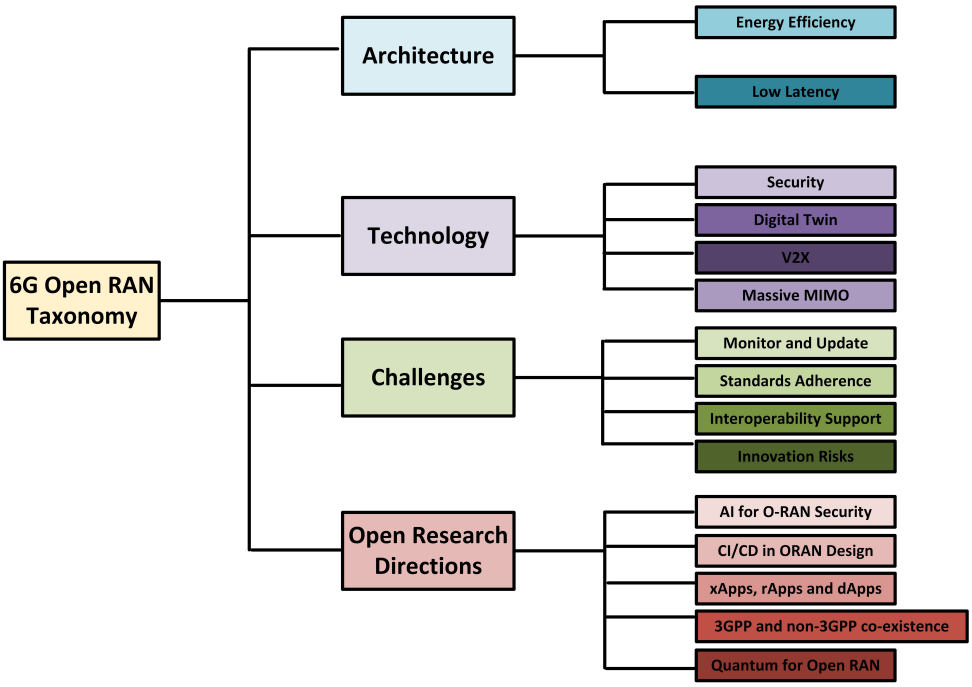


Figure 1. This Paper's Major Areas of Focus

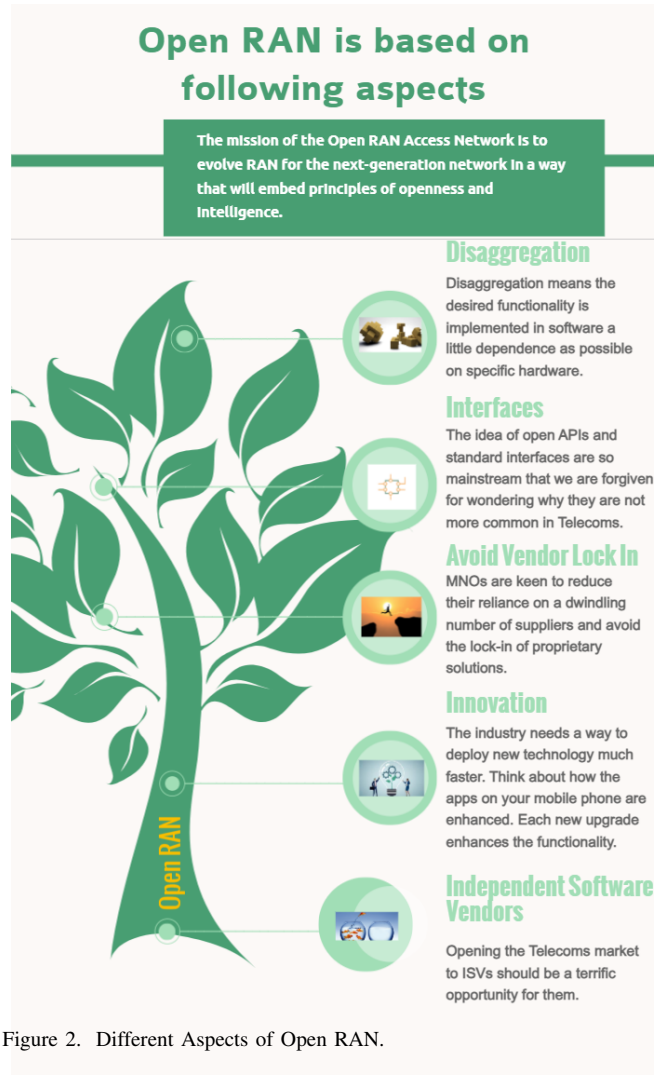


Figure 2. Different Aspects of Open RAN.

C. O-RAN Meets 6G

This paper provides a detailed review of schemes which sit at the intersection between O-RAN and 6G and were proposed to bridge the gap between them. The article starts with an extensive examination of how an O-RAN architecture instantiation and diverse proposed solutions provide performance-oriented support for innovative services, especially in relation to aspects related to energy efficiency (EE) and low latency (LL). Then, several recent technological rich avenues that use the O-RAN architecture are considered, including security, DT, vehicle-to-everything (V2X) and Massive Multi-Input Multi-Output (mMIMO). In each of these avenues, various solutions to communications-related problems are discussed, including those that involve beam-forming, traffic management, anomaly detection, channel modeling for RAN optimization and network traffic forecasting, as illustrated in Fig. 3.

D. 6G Challenges and O-RAN Solutions:

As the telecommunications industry moves towards the deployment of O-RAN in 6G networks, several challenges must be addressed to ensure its successful integration and widespread adoption. This sub-section explores the key challenges encountered in O-RAN research and deployment, focusing on issues such as interoperability, security vulnerabilities, performance optimization, and operational complexity.

EE: As 6G networks aim to support an exponentially higher number of connected devices and services, EE becomes critical. The need to reduce carbon footprints while maintaining high performance necessitates innovative approaches. O-RAN, with its disaggregated architecture, enables more efficient resource allocation and dynamic network management, which can significantly reduce energy consumption. Additionally, O-RAN's ability to integrate AI-driven energy-saving techniques offers a path towards sustainable 6G networks [16].

LL: 6G is expected to support ultra-low latency communications, essential for applications like real-time automation, remote surgery, and immersive experiences. Traditional RAN architectures may struggle to meet these stringent latency requirements. O-RAN, with its flexibility and ability to deploy virtualized network functions closer to the edge, can minimize latency by optimizing the data path and enabling faster processing at the network's edge [15].

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DT: The concept of DT, which involves creating real-time virtual replicas of physical systems, requires a network capable of handling massive amounts of data with minimal delay. 6G networks will rely heavily on DTs for applications in smart cities, industrial automation, and healthcare. O-RAN's modular design allows for the seamless integration of specialized functions that can efficiently manage and process the data streams needed to support DTs, ensuring accurate and timely virtual representations [17].

Security: With 6G expected to connect a vast array of devices, from personal gadgets to critical infrastructure, security becomes paramount. O-RAN introduces a more open and flexible architecture, which, while beneficial, also increases the attack surface. However, O-RAN's inherent ability to incorporate security functions at multiple layers of the network, combined with AI-driven threat detection, can address these security challenges. By enabling real-time monitoring and adaptive security measures, O-RAN can help safeguard the complex and diverse ecosystem that 6G will bring [18].

E. New Insights and Approaches Introduced by This Paper:

There are some survey articles that describe O-RAN and some related solutions, but none of them goes into great details on the architecture and how it is applied to innovative technologies and unique use cases. They do not identify any limitations either. In contrast, this survey article discusses O-RAN in details, the most relevant issues facing it, as well as potential remedies. This survey paper also offers a thorough overview of open research directions that might assist the research community in developing a coherent 6G framework based on O-RAN (6G O-RAN). The article has four key contributions:

1) While existing literature largely focuses on O-RAN's application in the context of 5G, this manuscript explores how O-RAN can be extended to support advanced 6G use cases. For example, the manuscript delves into the role of O-RAN in enabling anomaly detection and beam-forming.
2) The manuscript proposes a novel architectural framework that integrates O-RAN with emerging 6G technologies, such as DT. This framework outlines how O-RAN's disaggregated architecture can be adapted to support the deployment and management of these technologies, enabling realizing optimal decision-making for large scale physical systems and processes digitally represented in a cloud-based platform. This approach represents a significant advancement beyond the existing 5G-centric O-RAN architectures discussed in the current literature.
3) The manuscript presents original research findings that demonstrate the feasibility of using O-RAN for meeting specific 6G requirements. For instance, it includes architectures that showcase the potential of O-RAN in achieving ultra-low latency and high EE in a 6G context. These findings provide empirical evidence that O-RAN can be optimized for 6G's stringent demands, filling a gap in the existing body of work, which often lacks detailed performance analysis specific to 6G scenarios.
4) It introduces an innovative O-RAN Application Middleware Continuum, which illustrates the versatility of the O-RAN architecture-based solutions across various technologies and diverse use cases. It also includes a detailed survey on the challenges of O-RAN and proposed solutions. Finally, it presents a detailed list of open research directions, which can help frame the potential 6G O-RAN.

F. Paper Organization and Reading Map

The rest of the paper is organized as shown in Fig. 4. A brief tutorial on O-RAN is presented in Section II and existing surveys are discussed in Section III. Section IV presents the taxonomy which is at the centre of this survey. The O-RAN architecture and its EE and LL-related instantiations, along with solutions targeting specific novel use cases related to these avenues, are covered in details in Section V and VI, respectively. Section VII discusses various

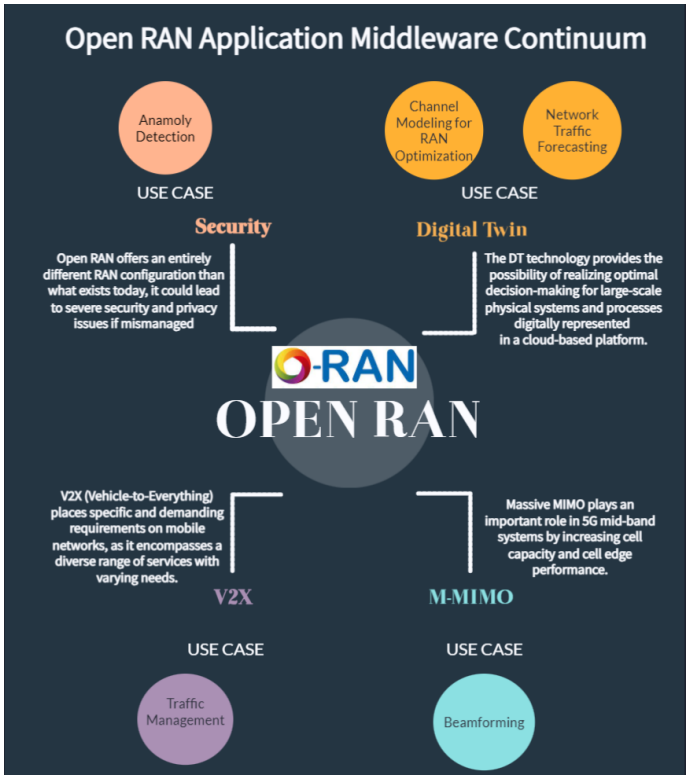


Figure 3. Novel O-RAN Application Middleware Continuum

innovative technologies and avenues that use the O-RAN architecture. Challenges in the efficient deployment of O-RAN are discussed in Section VIII and in section IX, some potential open research directions are presented. Finally, Section X concludes this survey paper and highlights major lessons learnt.

II. An O-RAN Tutorial

The internal interfaces within the single box that makes up the traditional RAN delivery method are closed and under the vendor's control. As we go toward O-RAN, the base station functions are separated into a centralized unit (CU), a distributed unit (DU), and a remote unit (RU) with open interfaces connecting them. Because of the open interfaces (such as Open Fronthaul) between them, multiple vendors can construct those entities using the O-RAN strategy. Furthermore, a crucial component is the RAN intelligent controller (RIC), which is separated from the processor units and enables the provision of management tasks such as self-organizing networking and radio resource management.

The claimed very important characteristics of the O-RAN i.e., (i) Dis-aggregation, (ii) Open ecosystem, (iii) Open interfaces, (iv) Decoupling hardware (HW) from software (SW), and (V) Intelligent management are shown in Fig. 5 and briefly discussed next.

1) **Dis-aggregation:** In order to enable the independent realization of each component, the RAN protocol stack is essentially divided into two halves by dis-aggregation. This seeks to address the issues of excessive energy consumption, high total cost of ownership, improved system performance through dynamic and intelligent radio resource management, and open, quick innovation across many component types while maintaining multi-vendor operability [8].
2) **Open Ecosystem:** A continuing change in mobile network topologies known as the "O-RAN ecosystem" allows service providers to use non-proprietary sub-components from several vendors. Table. I includes the major components of the diverse O-RAN ecosystem from the system integrator to chipset suppliers.

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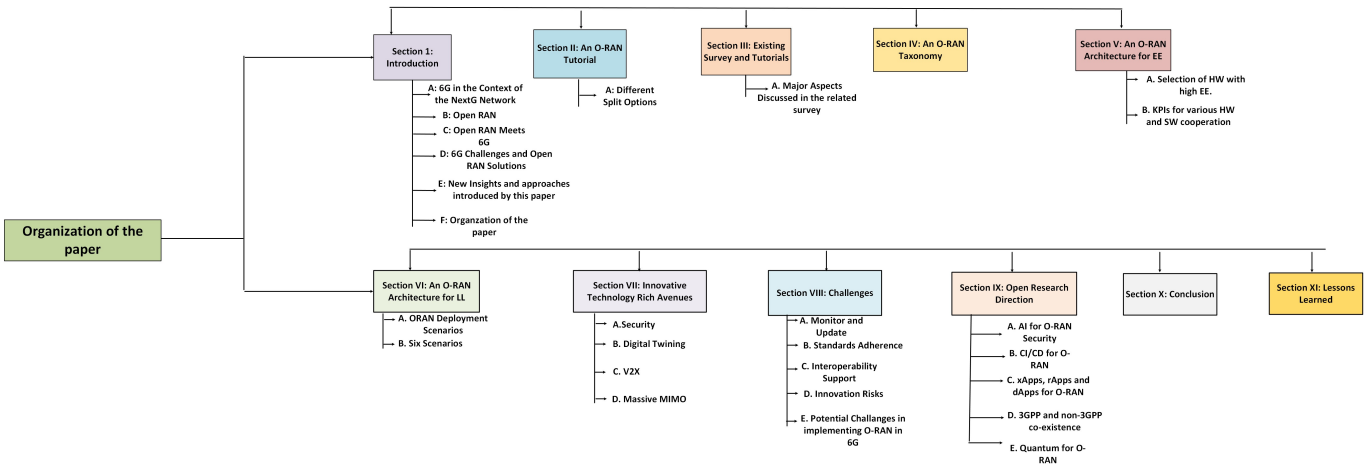


Figure 4. Organization of This Paper

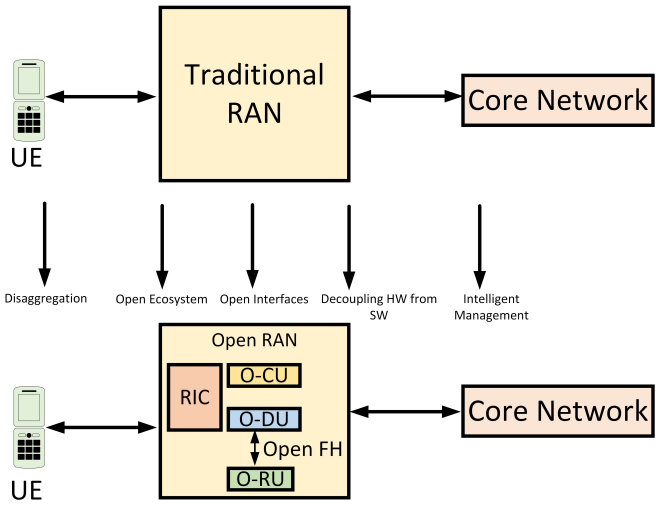


Figure 5. RAN Transformation from Traditional RAN to O-RAN based on Dis-aggregation, Open Ecosystem, Open Interfaces, Decoupling HW from SW and Intelligent Management.

3) **Open Interfaces:** There are several open interfaces that has been defined by both 3GPP and O-RAN alliance as shown in Fig. 6. The Interfaces marked in black are specified by 3GPP i.e., E1, F1-C, F1-U, NG-C and NG-U and those marked with blue are specified by the O-RAN Alliance i.e., A1, O1, O2, E2, M-Plane, and FH. The aim of this survey paper is not to explain in detail the interfaces, as [13] provides such explanations, but to highlight the most important ones.
   a) **A1 Interface:** is defined as the relationship between the Non-Real-Time RAN Intelligent Controller (Non-RT RIC) and the Near-Real-Time RAN Intelligent Controller (Near-RT RIC). Near-RT RIC gives Non-RT RIC policies enrichment information, and updates to their Machine Learning (ML) model, while Non-RT RIC responds with policy feedback.
   b) **E2 Interface:** touches and enters the designated base station entities, such as Open-DU and Open-CU. As a result, for example, we may monitor, suspend, override, control messages, and execute actions originating from xApps/Near-RT RIC, as well as obtain data gathering and feedback from those entities, to control what is occurring within that BS.
   c) **O1 and Open-Fronthaul M-plane interfaces:** a standard FCAPS (Fault, Configuration, Accounting, Performance, Security) interface that exchanges security, performance, monitoring, reconfiguration, and registration details with individual nodes, such as O-CU-UP (user plane), O-CU-CP (control plane), O-DU, O-RU, and Near-RT RIC.
   d) **O2:** to control the workload and platform resources (such as FCAPS for cloud computing platforms and resource scaling).
   e) **Service and Management Orchestration (SMO):** It is a framework responsible for the management and orchestration of RAN elements, supporting interfaces such as O1, A1, and O2. The SMO framework provides an integration fabric and data services for O-RAN managed functions, including the NE-RT RIC, O-CU, O-DU, and O-RU. Its key capabilities encompass FCAPS (Fault, Configuration, Accounting, Performance, Security) management, Non-Real Time RIC for RAN optimization, and O-Cloud Management, Orchestration, and Workflow Management.

   Three control loops are shown in Fig. 6. The first control loop is real-time and involves activities with a timescale of less than 10 ms, such as the O-DU scheduler. Next, there is the near-real-time control loop, wherein operations such as mobility management, interference management, and traffic steering take place, with timing ranging from 10 ms to 1 s. Ultimately, the outermost control loop integrates machine learning models together with orchestration and optimization functions to affect non-real-time operations lasting longer than a second.

4) **Decoupling HW from SW:** Conventional RAN organizations within the 4G Long-Term Advancement (LTE) era with D-RAN and C-RAN topologies were transcendentally brownfield, with firmly coupled equipment and program arrangements from a single merchant, coordinates through exclusive interfacing, as shown in Fig 7 (a) and (b). However, the reconfigurability and scalability of this traditional "black box" method are essentially unflexible, which are critical characteristics for the 5G network. Virtualization is the initial stage towards the development of the 5G RAN, converting the old RAN into a vRAN [9]. Using the concepts of network functions virtualization, a vRAN departs from the monolithic design of the traditional RAN by separating RAN application software from the underlying hardware platform [10], as shown in Fig. 7 (c). Instead of using proprietary, purpose-built hardware as in classic RAN, RAN applications within CU(s) and DU(s) are deployed as virtual network functions (VNFs) or cloud-native network functions (CNFs) running on COTS servers in vRAN. O-RAN further deconstructs vRAN into a software-defined, multi-vendor white-box solution that operates on commercialized-off-the-shelf (COTS) servers. Its open nodes, such as O-RAN O-RU, O-DU, and O-CU, have open and standardized interfaces and protocols between them [11], as shown in Fig 7(d).

5) **Intelligent Management:** RIC is built as a distinct entity within the O-RAN architecture and offers the capability to control the radio network and radio resources.

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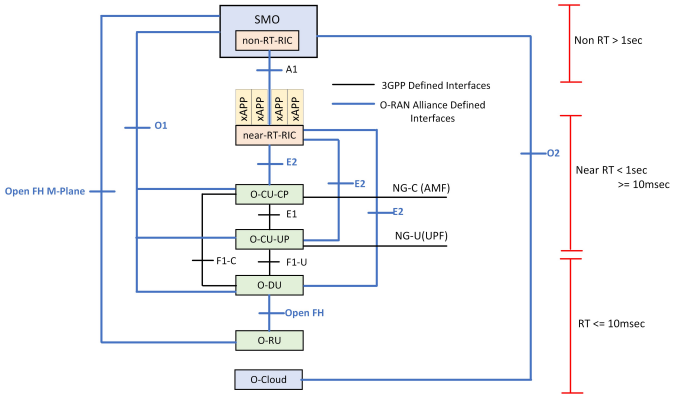


Figure 6. O-RAN Architecture with Open Interfaces defined by both 3GPP and O-RAN Alliance.

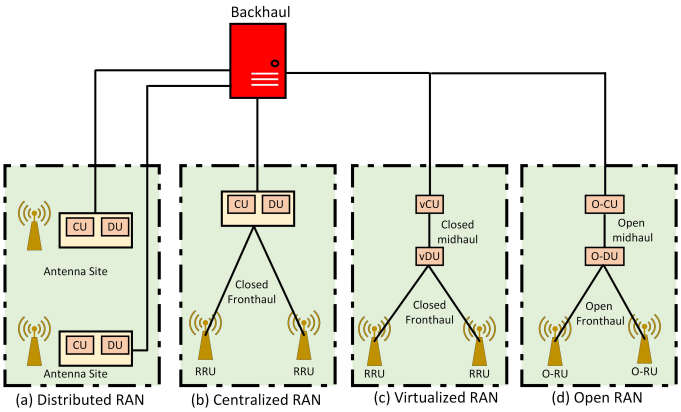


Figure 7. An illustration of RAN architecture options: D-RAN, C-RAN, vRAN and O-RAN.

a) **Near-RT RIC:** a logical node that provides fine-grained data collection and actions across E2 to enable near-RT control/optimization of RAN components and resources. AI/ML process may be included with RT RIC.
b) **Non-RT RIC:** a logical node that allows AI/ML process capture, policy-based guidance of Near-RT RIC applications and features, and Non-RT management and optimization of RAN elements and resources.

A. *Different Split Options*

In this section, we provide a detailed explanation of the functionalities of O-RU, O-DU, and O-CU, as well as various split options along with their respective advantages and disadvantages. To the best of authors' knowledge, this survey represents the only paper to comprehensively outline these split options.

Split options are crucial in 5G because they enable flexible and efficient deployment of network functions, optimizing performance, cost, and scalability. By allowing different parts of the radio access network (RAN) to be distributed across various locations, split options help balance processing workloads, reduce latency, and support diverse use cases such as ultra-reliable low-latency communications (URLLC) and massive machine-type communications (mMTC). These options are particularly relevant in 5G's disaggregated architecture, where functions can be centralized or distributed based on network demands, infrastructure availability, and operator strategies.

The DU and CU dis-aggregation, introduced by 3GPP as an evolution, involves O-CU supporting higher layers of the protocol stack like SDAP, PDCP, and RRC, while O-DU provides support for lower layers such as RLC, MAC, and PHY. The introduction of mid-haul enhances flexibility for transport options. Lower Layer Split (LLS) refers to the connection between O-RU and O-DU, known as

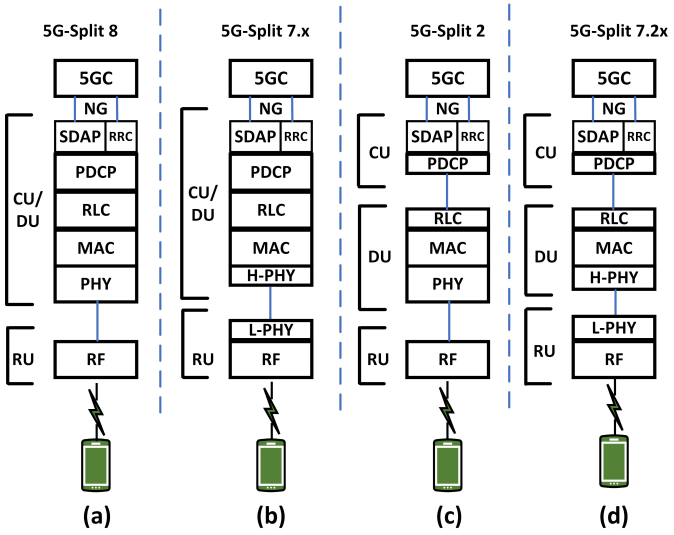


Figure 8. Multiple Split Options specified by O-RAN Alliance and 3GPP.

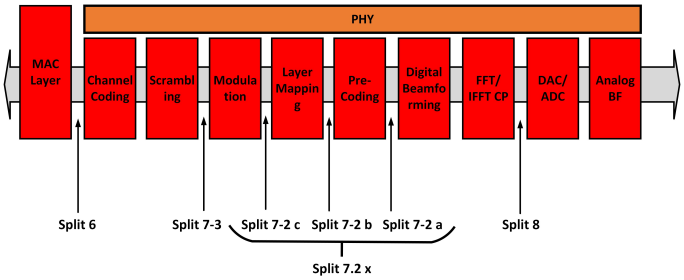


Figure 9. 'x' in Split 7.2x can either be 'a', 'b' or 'c', depending on the physical layer functionality, as illustrated.

front-haul, whereas Higher Layer Split (HLS) involves the connection between O-DU and O-CU, termed mid-haul. The O-RAN alliance has broadened the scope beyond the initial framework outlined by 3GPP to ensure interoperability among various hardware and software vendors. eCPRI serves as a standard enabling interoperability by allowing mixing and matching of radios from one vendor with DU/CU software from another, deployed on a COTS server. This approach supports the increasing demand for high bandwidth.

Various split options are presented in Fig.8, as specified by both the O-RAN alliance and 3GPP. In Fig.8 (a) illustrates the 5G-Split 8, similar to 4G-Split 8, where the CU/DU, analogous to the BBU, covers the complete protocol stack, and the RU manages the RF layer with the 5G CN. Conversely, Fig.8 (b) showcases the 5G-Split 7.x, also known as LLS, wherein the PHY layer functionality is split into H-PHY and L-PHY. In this Split, the CU/DU manages RRC, SDAP, PDCP, RLC, MAC, and H-PHY, while the RU handles L-PHY and RF. The "x" in this split can be either "a", "b" or "c" depending on whether some of the PHY functionalities reside in H-PHY or L-PHY (as shown in Fig. 9). Split 2, as shown in Fig 8 (c), also known as HLS, involves the splitting of CU and DU. In this arrangement, CU solely manages SDAP, RRC, and PDCP, whereas DU takes care of RLC, MAC, PHY, and RF. Split 7.2x, as depicted in Fig. 8 (d), combines Split 7.x and Split 2. Split 7.2x represents the most widely deployed O-RAN split. Subsequent sections will delve into the advantages and disadvantages of these different splits.

Table. II presents the advantages and disadvantages of Split 6 and Split 7-2a. Split 6, recommended by the Small Cell Forum, features CU consisting of RRC, SDAP, and PDCP, while DU comprises RLC and MAC, and RU handles PHY and RF. The FH interface is nFAPI, making it suitable for small cell deployment in rural areas. Conversely, Split 7-2a is better suited for macrocell deployment in urban areas with FH interface as eCPRI. Split 6 (nFAPI) is bandwidth-efficient but struggles with advanced MIMO techniques, making it

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Table I
O-RAN ECOSYSTEM [12]

O-RAN Ecosystem	
System Integrators	Cisco, IBM, Rakuten, NOKIA, NEC, SIEMENS
vRAN Software Suppliers	Meta, Juniper Networks, Parallel Wireless, Radisys, Capegemini, Mavenir
RU HW and SW Suppliers	NEC, Benetel, Samsung, Baicells, Toshiba, Accelleran
CU and DU HW Suppliers	DELL, HP, Intel, Samsung, Accelleran, Intel
Secure Virtualization Platform Suppliers	vmware, IBM, aws, Microsoft, Kesight, ZTE
Chipset Suppliers	Nvidia, Qualcomm. Samsung, Xilinx, intel, AMD

Table II
COMPARISON BETWEEN SPLIT 6 AND SPLIT 7-2X

Split Type	FH Interface	Advantages	Disadvantages
Split 6	**FH Interface is nFAPIfrom Small Cell Forum**	**FH BW will be lower as more functions are in O-RU.**	**nFAPI or split 6 approach is challenged in supporting massive MIMO functionality that is viewed as key enabling technology for 5G deployments. Only MAC PDUs is transmitted, not IQ data.**
		Enable the PNF implementing the small cell RF and PHY layer to be remotely located from the VNF implementing the small cell MAC layer and upper layer RAN protocols.	**However, mMIMO is more applicable to outdoor macro cellular coverage, where it can be used to handle high mobility and suppress cell-edge interference use cases. Hence, there may be a subset of 5G deployments where mMIMO support is not rqquired.**
		the nFAPI transport BW is able to vary according to served traffic, enabling statistical multiplexing to be used over the FH IP network.	**When considering other overheads, the analysis indicated that the difference in DL BW between Split 7-3 and Split 7-2x with modulation compression was estimated to be around 7%.**
		Contrast with the alternative CPRI split, also referred to as "Split 8", that required BW expansion up to 30 fold and a constant bit rate connection, even if there is no traffic being served in a cell.	**Putting a lot of functions in O-RU make it very complex and it will consume a lot of power for processing and it will be not be very interoperable.**
Split 7-2a	**FH interface is eCPRI**	**When compared to Split 6, the O-RAN lower layer split 7-2x support all advanced RF combining techniques, including higher-order MIMO.**	**The use of frequency domain IQ symbols can lead to a significant increase in the FH BW when compared to the original transport channels.**
		Instead of supporting individual transport channels over the nFAPI interface, Split 7-2a defines the transport of frequency-domain IQ-defined spatial across the lower layer FH interface.	**the BW expansion due to split 7-2 occuring "below" the modulation function, where the original bits to be transmitted are expanded to over 18 bits after 16 QAM modulation, even when using a block floating point compression scheme.**
		Implementing the channel coding above the RU in Split 7-2a is a nominal in BW, the benefit to such an approach is the significant simplification of the RU by removing the need to perform channel decoding.	**When considering other overheads, the analysis indicated that the difference in DL BW between split 7-3 and Split 7-2x with modulation compression was estimated to be around 7%.**
		Performing resource element mapping/demapping on the O-DU side, data will be transmitted after user multiplexing thereby simplifying control signals on the FH and making it easier to achieve multi-vendor RAN.	**HARQ delay constraints.**

suitable for small cells rather than large macro deployments. Split 7-2a (eCPRI) supports advanced RF processing and multi-vendor interoperability but demands significantly higher fronthaul bandwidth and stricter delay requirements.

III. EXISTING SURVEYS AND TUTORIALS

This section summarizes key themes and contributions from previous surveys on O-RAN, highlighting the specific aspects they address, their limitations, and how they relate to our study. Table III presents a comparative analysis of various survey papers related to O-RAN and its role in the evolution of next-generation wireless communication networks, particularly 6G. It introduces each paper based on their list of authors (e.g., Azariah et al., Wypiór et al.) with corresponding reference numbers for citations, then evaluates the articles based on different aspects, which include: **Key Problems Addressed** such as: RAN evolution history, O-RAN architecture principles, research gaps in security, privacy, and quantum communications and technical challenges like mMIMO and DT integration and focus area. The table is divided into key columns on: **Architecture:** EE and LL. **Key Technologies:** DT, V2X, mMIMO, Security. **Challenges:** e.g. AI for Security and **New Research Directions:** CI/CD, dApps, Co-existence, Quantum Technology. '✓' indicates that the paper extensively covers the topic corresponding to the current column, whereas 'x' means that the topic is either just mentioned but not deeply explored or it is not addressed. *"This article"* refers to the current article, highlighting its broader coverage compared to the previous works, focusing on all the indicated aspects EE, LL, Security, etc.

Several tutorials have been published to formally introduce O-RAN, its architecture, novel use cases and their related challenge. In [9], Azariah *et al.* provided a thorough introduction to O-RAN and its significance, spoke about its difficulties and possible avenues for future study, and addressed some of the difficulties from an open-

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source standpoint. They have started by summarizing the RAN's progression from traditional to vRAN. It aids in recognizing the variations and advancements in RAN technology. Subsequently, the O-RAN movement and its associated technologies were presented. Following that, they talked on O-RAN-related initiatives, events, and standards. They then gave a brief explanation of the O-RAN Alliance's standardized O-RAN architecture and its application cases. Following that, they spoke about the difficulties and potential lines of inquiry for each standard architecture component of an O-RAN solution. While the article helped the readers grasp the issues encountered by O-RAN, it did not go into great details on the architectures, technologies, or future research directions, even though it provided an excellent resource for fundamental knowledge. In [12], Wypior *et al.* outline the progression of RAN toward open models and attempt to identify market trends and possible advantages of open RANs.The same flaw as mentioned for [9] also affects this work. This research just provides us with rudimentary understanding of open RAN and associated issues.

Polese *et al.* provided the first comprehensive O-RAN lesson in [13]. They also go over the primary research obstacles and preliminary findings. They gave a thorough explanation of the O-RAN specifications, including the interfaces, architecture, and design tenets. They go on to explain how 3GPP-defined RANs may be efficiently managed and controlled through the use of O-RAN RICs. Based on this, they talk about the advancements and difficulties with O-RAN networks, such as the security and standardization concerns, as well as the Artificial Intelligence (AI) and ML processes that are made possible by the architecture and interfaces. Lastly, they discuss current research findings, experimental research platforms that may be used to build and test O-RAN networks, and future plans for O-RAN development. Overall, this survey study covers every facet of open-ran network design, challenges, and future research directions; however, it omits discussion of technologies that leverage O-RAN network architecture to address novel use cases. Ali *et al.* comprehensive study and imaginative description of Quantum Computing (QC) are presented in [14]. QC has the ability to reduce the computing complexity of upcoming 6G communication systems. Furthermore, a thorough discussion of the relevance of quantum repeaters, quantum interconnects, and quantum security has been held; these topics are seen to be extremely helpful and significant in the advancement of Quantum Information Technology. According to this study, quantum technology is still developing gradually, and creating safe and effective quantum communication equipment for 6G communication networks may present some difficulties for 6G communications enabled by QC. The report also outlines the possible obstacles and opportunities that a quantum-enabled 6G network may face in the future. While this study does not fully align with our survey report, it does provide us with a valuable avenue for further research on the application of QC in O-RAN.

In [15], Singh *et al.* provided an explanation of the history of O-RAN, its reference architecture, and the evolution of RAN. This paper presents an architecture that represents a step toward a software-oriented network. They also talked about a number of difficulties in implementing O-RAN. Similar to [9], [12], this study has the same shortcomings in that it only discusses the background of O-RAN, a thorough comprehension of it, and the difficulties associated with deploying O-RAN. Abubaker *et al.* examined the O-RAN from an EE standpoint in [16]. They did this by examining the most recent power consumption models and EE strategies that have been put out to reduce the O-RAN's energy usage. Furthermore, the difficulties in optimizing the EE of O-RAN as well as potential directions for further study are also emphasized. A portion of our survey paper is covered in this publication. They aid in our comprehension of the O-RAN's EE design, but they do not discuss how other technologies may use that architecture to address cutting-edge use cases. In [17], Masaracchia *et al.* want to give a thorough understanding of how O-RAN and DT are two complimentary avenues that, when combined, will allow for the deployment of an intelligent and durable 6G RAN. DT will be especially important since it will strengthen the intelligence, autonomy, and openness pillars that form the foundation of O-RAN. The principles of DT and O-RAN are briefly reviewed. Potential use cases and services that may be provided by an O-RAN architecture based on DT are then demonstrated. Future research directions and challenges are also emphasized and spoken about. Similar to [17], this work does not quite align with our survey, but it does provide insight into how DT, a cutting-edge technique, may be integrated with O-RAN to handle a few unique use cases.

Polese *et al.* in [18] have examined the role of the O-RAN paradigm as a crucial facilitator of advancements in 6G networks, focusing on both system-level and architectural aspects. They have analyzed essential criteria for 6G, such as EE, coverage, resilience, cost-effectiveness, complexity, and innovation, and investigated the alignment of O-RAN design principles with these requirements. For each principle, they have identified the outstanding challenges hindering its complete realization and integration into commercial networks. This study builds upon [13] and provides more insight into the interplay between 6G and O-RAN. However, LL and High Throughput (HT) design, as well as several innovative use cases, were not covered. Liyanage *et al.* as detailed in [19], examine the security and privacy threats and obstacles linked with the O-RAN architecture. Subsequently, potential security and privacy measures are deliberated to safeguard the O-RAN architecture, alongside an overview of pertinent security standardization endeavors related to the O-RAN security. Lastly, considerations are given to leveraging O-RAN for implementing enhanced security and privacy solutions within 5G and future RAN environments. Since this study is entirely focused on security, we did not identify any restrictions in comparison to our survey paper. This survey paper covers the security component that is also included in our survey paper.

Regrettably, as presented in Table III, the previously mentioned tutorials and surveys lack a critical evaluation of each assessed contribution, utilizing clearly defined and justified criteria.

A. Major Aspects Discussed in Related Surveys

Understanding the evolution of O-RAN requires a detailed examination of prior surveys that have explored its architecture, technologies, and challenges. This section identifies the key aspects discussed in related literature, highlighting the primary research themes, gaps, and how they align with our study. By categorizing these aspects—including security, energy efficiency, quantum computing, digital twins, and V2X communications—we aim to provide a structured analysis of existing contributions while identifying areas requiring further exploration. Each aspect is supported with references to relevant studies, offering a comprehensive overview of the current state of O-RAN research.

1) **V2X [67]**: V2X communication is crucial for the development of smart transportation systems and autonomous vehicles, and O-RAN is poised to play a key role in next-generation cellular networks (5G and beyond), which will support such technologies. Since O-RAN provides greater flexibility, programmability, and cost efficiency for 5G deployments, its architecture can facilitate optimized V2X solutions. Therefore, V2X is relevant to O-RAN in terms of ensuring low-latency and reliable communication channels for vehicular applications.
2) **EE [16]**: O-RAN is highly customizable, allowing operators to optimize network resources dynamically. EE is a key consideration for both cost reduction and sustainability. O-RAN can reduce energy consumption through efficient resource allocation, dynamic sleep modes, and by running energy-intensive tasks in centralized or cloud-based locations. This is particularly important in the context of green communication and future network rollouts.
3) **Security [19]**: Given that O-RAN leverages open interfaces and relies on a multi-vendor ecosystem, security is paramount. The openness of the system increases the risk of cyberattacks, such as network intrusion, data interception, or attacks on infrastructure. Proper security measures, including authentication, encryption, and intrusion detection systems, are needed to protect

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the network. Therefore, examining how each survey addresses security is highly relevant to understanding the maturity of O-RAN.

4) **DT [17]**: A DT is a virtual representation of a physical system or process, and in the context of O-RAN, it can be used to simulate and optimize network operations. DTs enable predictive analysis and real-time monitoring of the network, which is useful for performance optimization, fault detection, and network planning. This technology can help operators maintain optimal network conditions in a dynamic and distributed O-RAN architecture.
5) **mMIMO [74]**: O-RAN is designed to support advanced radio technologies such as mMIMO, which is essential for achieving high spectral efficiency and data throughput in 5G networks. mMIMO is critical in meeting the demands for bandwidth in densely populated areas and improving network capacity. The flexibility of O-RAN makes it well-suited to incorporate and manage mMIMO systems efficiently.

Each of the categories described above directly relates to the architecture, technologies, and challenges of O-RAN. Here is how:

1) Quantum and V2X represent emerging technologies that O-RAN may need to accommodate as 5G and 6G networks evolve.
2) EE, LL, and security are foundational for the practical deployment of O-RAN systems, ensuring that they meet performance and sustainability goals.
3) DTs align with the programmability and virtualized nature of O-RAN, allowing for smarter, automated network operations.
4) mMIMO is a key enabler for 5G performance, and its integration into O-RAN is vital for achieving high capacity and spectral efficiency.

Each of these criteria is essential in understanding how O-RAN meets the demands of modern wireless communication networks and how it will evolve.

The categories under "Challenges" and "New Research Directions" in the table represent critical obstacles that need to be addressed for widespread O-RAN adoption, as well as future areas of research that could further enhance O-RAN capabilities.

Challenges: While O-RAN presents a promising approach for next-generation networks, its deployment and adoption face several key challenges. Challenges must be addressed to ensure seamless integration, efficiency, and sustainability in 6G networks. By fostering collaboration between industry stakeholders, investing in automation and AI-driven management tools, and strengthening standardization efforts, O-RAN can overcome barriers. Addressing challenges will be essential to unlocking the full potential of O-RAN in transforming 6G networks. We have identified four different challenges, and they have explained in detail in the subsequent sections.

New Research Directions:

Quantum [14]: While quantum technology might seem unrelated to O-RAN at first glance, it can have significant implications for secure communication. As O-RAN involves disaggregated components across the Radio Access Network (RAN) with extensive use of cloud technologies, quantum communication could potentially enhance security protocols in the future. For example, quantum cryptography can ensure unbreakable encryption methods for data transfer in distributed networks like O-RAN. The inclusion of quantum technology hints at future security and performance improvements. The connection can be justified through several architectural enablers:

AI/ML-Driven Optimization: O-RAN architectures heavily rely on AI/ML algorithms for dynamic spectrum management, resource allocation, and real-time decision-making. QC can accelerate complex optimization problems that traditional processors struggle with, such as beamforming algorithms, large-scale network optimization, and real-time data analytics within the RIC.

Quantum Cryptography for Enhanced Security: As O-RAN introduces more open interfaces, security risks increase. Quantum Key Distribution (QKD) offers unbreakable encryption, ensuring secure data exchange across these interfaces, especially between the DU, CU, and RIC.

Advanced Channel Modelling: QC can process large datasets to model wireless channels more accurately, enhancing performance in scenarios involving massive MIMO and ultra-dense networks, which are integral to 6G.

CI/CD: CI/CD is important for rolling out updates and improvements in O-RAN environments. CI/CD practices must be carefully integrated to ensure that software changes do not disrupt the network or compromise security. CI/CD necessity arises from key architectural aspects: **Cloud-Native and Virtualization Principles:** O-RAN disaggregates traditional RAN functions into microservices running in containers or virtual machines. CI/CD pipelines are critical for automating the development, testing, and deployment of these microservices, ensuring rapid updates without service disruption. **xApps and rApps Lifecycle Management:** The RIC enables third-party applications (xApps for near-real-time functions and rApps for non-real-time functions). CI/CD facilitates continuous delivery of updates to these applications, enhancing flexibility and adaptability in network operations. **Automation and Orchestration:** O-RAN's Service Management and Orchestration (SMO) framework benefits from CI/CD to automate configuration, scaling, and monitoring of network functions, ensuring high availability and reduced operational costs.

dApps: The use of dApps is a novel area of research for O-RAN, as it introduces new possibilities for leveraging blockchain technology for secure and transparent network management. This could revolutionize the way operators deploy, manage, and secure the RAN.

Coexistence: Another emerging research area is how O-RAN will coexist with traditional RAN architectures and legacy systems. As networks transition to 5G and eventually to 6G, ensuring compatibility and smooth operation across different network types is an important research focus.

Additionally, they overlook conducting a thorough examination of the literature. Notably absent is also the consideration of various open RAN architectures and their applicability across different technologies for addressing innovative use cases. In contrast, this survey article provides a more detailed and focused analysis of O-RAN in the context of 6G, offering several novel contributions:

1) **Detailed Architectural Analysis:** Unlike previous surveys that provide a general overview, this survey offers a comprehensive analysis of O-RAN architecture specifically optimized for 6G network performance, with an emphasis on EE, LL, and HT. The depth of analysis includes a thorough examination of various split options, hardware-software decoupling, and the implications for 6G networks.
2) **Innovative Use Cases:** This survey introduces the concept of an O-RAN Application Middleware Continuum, illustrating how O-RAN architectures can be applied across various emerging technologies such as DT, V2X communications, and mMIMO. Previous surveys have not explored these applications in the context of O-RAN with such detail, particularly within the 6G framework.
3) **Comprehensive Challenge Identification:** We provide a detailed survey on the challenges of deploying O-RAN, particularly in the context of 6G, and propose potential solutions. This includes discussions on interoperability, security, and the integration of AI/ML techniques. While other surveys have touched on these issues, our work goes further by offering concrete solutions and identifying open research directions that are crucial for the evolution of 6G O-RAN.
4) **Open Research Directions:** The survey concludes by outlining specific open research directions that have not been comprehensively addressed in the literature. These include the integration of quantum computing for O-RAN, CI/CD for O-RAN, and the development of secure and efficient xApps, rApps, and dApps. These directions are critical for advancing the state of the art and are discussed in detail to provide guidance for future research.

This survey distinguishes itself from existing literature by not only summarizing the state of the art but also by offering new insights

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Table III
MAJOR O-RAN RELATED SURVEYS

Survey Paper	Key Problems Addressed	Architecture		Key Technologies				Challenges	New Research Directions				
		EE	LL	Security	DT	V2X	mMIMO		AI for Security	CI/CD	dApps	co-existence	Quantum
Azariah et. al. [9],	Provides a succinct overview of the history of RAN evolution as well as the cutting-edge technologies used in O-RAN.	x	x	x	x	x	x	✓	x	x	x	x	x
Wypiór et al., [12]	Outlines the evolution of RAN in the direction of open models, tries to highlight possible advantages of O-RAN and describes market development.	x	x	x	x	x	x	✓	x	x	x	x	x
Polese et. al, [13]	Provides the first comprehensive O-RAN lesson, including primary research obstacles and preliminary findings.	x	x	✓	x	x	x	✓	✓	x	x	x	x
Ali et.al, [14]	Shows that quantum technology is still developing gradually, creating effective and safe quantum communication solutions and equipment for 6G networks.	x	x	✓	x	x	x	✓	x	x	x	x	✓
Singh et.al, [15]	Discusses the development of RAN as well as the potential design and characteristics of O-RAN.	x	x	x	x	x	x	✓	x	x	x	x	x
Abubakar et.al., [16]	Provides a succinct overview of the history of RAN evolution as well as cutting-edge technologies used in O-RAN.	✓	x	x	x	x	x	✓	x	x	x	x	x
Masaracchia et.al, [17]	The principles of DT and O-RAN are briefly reviewed.	x	x	x	✓	x	x	✓	x	x	x	x	x
Polese et.al, [18]	Examines how O-RAN design concepts relate to the main 6G criteria.	✓	x	x	x	x	x	✓	x	x	x	x	x
Liyanage et.al, [19]	Analyzes the security and privacy risks and challenges associated with the O-RAN architecture.	x	x	✓	x	x	x	✓	✓	x	x	x	x
This article	Looks at the O-RAN architecture from the perspective of EE, HT and LL and how this architecture addresses novel use cases; presents challenges and novel potential research directions.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

and research directions specifically tailored for 6G networks. We provide a detailed architectural analysis, introduce novel use cases that have not been explored in previous surveys, and propose solutions to the unique challenges of deploying O-RAN in a 6G context. These contributions make this survey a critical resource for researchers and practitioners looking to advance the field of O-RAN and 6G. There are numerous other surveys [94]–[109] and studies that are closely related to O-RAN but fall outside the specific scope of this survey. These works address various aspects of O-RAN, such as advanced architectural designs, detailed protocol analyses, cutting-edge deployment strategies, and other specialized research directions. While we do not delve into these topics in this survey, they provide valuable insights and complement the areas discussed here. To offer readers a broader perspective and to guide them toward additional research avenues, we have included citations to these studies in our paper. These references can serve as a starting point for readers interested in exploring more specialized or adjacent topics related to O-RAN, enabling them to gain a deeper understanding of the field and its diverse dimensions.

[94] analyses the architecture and performance of the near real-time RIC, emphasizing its operations and benchmarking techniques. It also highlights the critical role of real-time data in ensuring efficient network management. [96] discusses Intel-based solutions for implementing Cloud-RAN architectures, focusing on their potential to reduce costs and improve computational efficiency for 5G and 6G networks. [101] proposes the use of the Extended Berkeley Packet Filter (eBPF) as a tool for enhancing observability, networking, and security in cloud-native environments for 5G and 6G. [104] introduces OPlaceRAN, an orchestration framework for the virtualized RAN ecosystem, focusing on optimizing resource placement for efficiency. [108] discusses the design and implementation of network DTs tailored for RAN, emphasizing their utility in planning, management, and troubleshooting. [109] reinforces the applicability of eBPF for enhancing networking and security features for O-RAN deployments in next-generation networks.

IV. An O-RAN Taxonomy

This survey introduces a classification approach for the most recent O-RAN contributions in terms of the architecture and technology, and discusses related challenges and open research directions in 5G/6G networks. The innovative taxonomy described could be considered a foundational resource for understanding key design elements and evaluating proposed algorithms, including their strengths and weaknesses. Given the diverse literature on O-RAN, organizing relevant works in a systematic manner is challenging. The proposed taxonomy, outlined with four distinct branches, is depicted in Fig. 1. We identify four main categories; (1) Architecture, (2) Technologies, (3) Challenges, and (4) Open Research Directions. Conducting a literature review from these four perspectives is a logical decision, as the majority of researchers in this field tend to focus on one of these aspects. Within the initial category, known as "architectures" for addressing various architectures in O-RAN, three distinct architectural proposals have been put forward:

(a) EE: In terms of EE, the O-RAN architecture holds significant potential for several advantages:

1) **Virtualization and Software Optimization:** O-RAN facilitates the virtualization of network functions, allowing them to run on general-purpose hardware. This approach, coupled with software optimization, can enhance resource utilization and decrease energy consumption compared to traditional dedicated hardware setups.
2) **Multi-Vendor Competition:** The interoperability enabled by O-RAN encourages competition among equipment vendors. This competition incentivises the development of more energy-efficient solutions to maintain competitiveness in the market, ultimately driving innovation and efficiency improvements.
3) **Dynamic Resource Allocation:** O-RAN's flexibility permits dynamic resource allocation based on varying traffic demand and network conditions. Through intelligent resource management, operators can optimize energy consumption and minimize wastage.
4) **Integration of Green Networking Technologies:** An open RAN architecture can integrate green networking technologies, such as energy-efficient hardware components, renewable energy sources, and intelligent power management systems. These technologies further augment EE within the network infrastructure.
5) **Network Optimization and Load Balancing:** Advanced optimization and load-balancing algorithms inherent in O-RAN architectures aim to maximize network efficiency while minimizing energy usage. These algorithms dynamically adjust network parameters to optimize energy consumption without compromising network performance.

In conclusion, despite the initial investment and restructuring required for transitioning to an open RAN architecture, its potential to enhance EE, reduce operational costs, and stimulate innovation renders it an appealing option for operators seeking to establish sustainable and future-ready cellular networks.

(b) Throughput: The architecture within an O-RAN setup typically encompasses several crucial components and technologies:

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1) **Baseband Processing:** This involves handling baseband signals, encompassing tasks like modulation/demodulation, encoding/decoding, and other signal processing functions. In O-RAN, this processing can be executed using software-defined radio (SDR) platforms operating on standard servers or specialized hardware provided by various vendors.
2) **Centralized and Distributed Units:** Unlike traditional RAN architectures where baseband processing is usually centralized in a base station, O-RAN allows for distributed processing across different units. This may involve a CU responsible for tasks such as scheduling, coordination, and management, as well as DUs located nearer to the radio antennas for Radio Frequency (RF) processing.
3) **Open Interfaces:** A fundamental aspect of O-RAN is the utilization of open interfaces between different RAN components. These interfaces, defined by organizations like the O-RAN Alliance and Telecom Infra Project, facilitate interoperability between hardware and software components from diverse vendors, promoting a more modular and adaptable RAN architecture.
4) **Virtualization and Orchestration:** O-RAN architectures frequently leverage virtualization technologies like NFV and cloud-native architectures. This allows RAN functions to operate as software instances on standard hardware infrastructure, enhancing flexibility and scalability. Orchestration platforms are employed to automate the deployment and management of virtualized RAN functions.
5) **Radio Units (RUs):** These are the physical radio transceivers responsible for transmitting and receiving RF signals to and from mobile devices. In O-RAN, RUs from various vendors should seamlessly interoperate with the rest of the RAN components via open interfaces.

Overall, the throughput architecture in O-RAN strives to deliver high performance, flexibility, and cost-effectiveness through the adoption of open standards, virtualization, and distributed processing. This architecture empowers mobile operators to deploy and manage their networks more efficiently while nurturing innovation and competition among RAN vendors.

(c) LL: The O-RAN architecture can be customized to meet low-latency demands through several key strategies:

1) **Edge Computing:** Integration of edge computing capabilities brings compute resources closer to end-users at the network edge. This facilitates local data processing and application execution, leading to reduced round-trip times for latency-sensitive services.
2) **Real-time Processing:** Prioritizing real-time processing within O-RAN systems, especially for latency-sensitive applications like online gaming or industrial automation, minimizes processing delays within the network, thereby reducing latency.
3) **Network Slicing:** O-RAN enables network slicing, allocating virtualized network functions to specific slices optimized for different use cases. Resources can be dedicated to low-latency slices, ensuring that critical applications receive adequate network resources to meet latency requirements.
4) **Dynamic Spectrum Allocation:** Dynamic spectrum allocation techniques prioritize low-latency traffic over the air interface, dynamically allocating resources based on application needs and network conditions to provide preferential treatment to latency-sensitive services.
5) **Quality of Service (QoS) Management:** Sophisticated QoS management mechanisms can be implemented in O-RAN systems to prioritize traffic based on latency requirements. Enforcing strict QoS policies and employing traffic shaping techniques ensures that latency-sensitive applications receive necessary network resources to meet performance objectives.
6) **Multi-access Edge Computing (MEC):** Leveraging MEC platforms, O-RAN brings compute and storage resources closer to end-users, reducing transit times to centralized data centers and benefiting latency-sensitive applications.
7) **Optimized RAN Design:** O-RAN architectures can optimize RAN design to minimize air interface latency. Techniques such as beam-forming, interference mitigation, and advanced modulation schemes enhance spectral efficiency and decrease round-trip times.

By integrating these strategies, open RAN architectures can effectively cater to low-latency requirements across various applications, empowering operators to deliver high-performance services while optimizing network infrastructure efficiency.

The second category of the proposed ontology includes different technologies for which open ran can be adopted by to address open challenges. Four major sub-categories have been identified:

Security is a paramount concern in the design of the O-RAN architecture, particularly given its distributed and dis-aggregated nature. There are several considerations and approaches to ensuring security in O-RAN deployments like Standardized Security Protocols, Authentication and Access Control, Encryption and Regulatory Compliance. By addressing these security considerations and implementing robust security measures, operators can enhance the resilience and trustworthiness of O-RAN deployments, providing secure and reliable communication services for users. Ongoing vigilance, collaboration, and investment in security are essential to effectively mitigate evolving threats and vulnerabilities in O-RAN environments.

Integrating O-RAN with **DT** technology presents numerous advantages across diverse industries, especially in terms of improving operational efficiency, optimizing network performance, and facilitating new use cases. This integration empowers operators with robust tools to enhance network performance, streamline operations, and foster innovation in various sectors. Leveraging real-time data analytics, predictive modeling, and simulation capabilities enables operators to unearth valuable insights and opportunities for value creation within their network deployments.

The integration of O-RAN with the **V2X** communications technology, offers significant potential for advancing connected vehicle capabilities and enhancing road safety. V2X communications include various technologies facilitating Vehicle-to-Vehicle (V2V), Vehicle-to-Infrastructure (V2I), Vehicle-to-Pedestrian (V2P), and Vehicle-to-Network (V2N) interactions. By harnessing the capabilities of the O-RAN architecture, operators can fully leverage V2X communications to improve road safety, alleviate traffic congestion, and introduce new mobility services in connected and automated vehicle environments. This integration enables seamless communication between vehicles, roadside infrastructure, pedestrians, and the network, fostering a safer and more efficient transportation ecosystem. Through O-RAN, operators can deploy flexible and scalable communication networks that adapt to dynamic traffic conditions and evolving connectivity requirements. This agility facilitates the rapid deployment of V2X services and the optimization of communication resources to prioritize safety-critical applications.

The O-RAN architecture presents a compelling opportunity for deploying **mMIMO** systems, offering a host of advantages in terms of flexibility, scalability, and cost-effectiveness. mMIMO technology, which utilizes a large number of antennas to enhance spectral efficiency and capacity, can significantly boost network performance when integrated with O-RAN. The flexibility of Open RAN architecture allows operators to deploy mMIMO antennas in a more agile and dynamic manner. With O-RAN's disaggregated approach, operators can mix and match hardware and software components from different vendors, selecting the best-in-class solutions for their specific needs. This flexibility enables operators to adapt their network infrastructure to evolving traffic patterns and changing user demands. Moreover, O-RAN's scalability is particularly advantageous for mMIMO deployments. Operators can easily scale up the number of antennas and baseband processing resources to accommodate growing capacity requirements without overhauling the entire network architecture. This scalability ensures that operators can efficiently meet the increasing demand for high-speed data services without incurring excessive costs or complexity.

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The third category of the proposed ontology includes **challenges** for O-RAN. Four major challenges have been identified: Examining Again, Standards Adherence, Sustaining Independence, and over innovation provides insight into both the opportunities and potential obstacles facing its widespread adoption: (a) Examining Again: Complexity Management and Optimization , (b) Standards Adherence: Interoperability and Evolution of Standards, (b) Sustaining Independence: Vendor Lock-in Concerns and Integration Complexity, (c) Over Innovation: Fragmentation and Complexity and Rapid pace of development. To tackle these challenges effectively, industry stakeholders, including operators, vendors, standardization bodies, and regulatory authorities, must collaborate closely.

The fourth category of the proposed ontology includes **open research direction** for O-RAN. The following five major open research directions have been identified:

1) **AI for O-RAN Security:** Leveraging AI in O-RAN security can improve the detection of threats, anomalies, and response capabilities, thus bolstering the network's overall security stance. Integrating AI into security strategies for O-RAN can elevate threat detection, response, and resilience, empowering operators to more effectively safeguard their network infrastructure, sensitive information, and crucial assets from evolving cyber threats. Nonetheless, it is imperative to verify that AI-driven security solutions are appropriately configured, validated, and seamlessly integrated into existing security frameworks to optimize their efficacy while mitigating potential risks.
2) **Continuous Integration and Continuous Deployment (CI/CD) for O-RAN:** CI/CD practices are pivotal in the development, testing, and deployment of O-RAN solutions, facilitating operators and vendors to furnish dependable, scalable, and feature-rich network infrastructure. CI/CD methodologies expedite the progress of O-RAN solutions, enhancing agility, dependability, and scalability, and nurturing a collaborative environment that promotes ongoing enhancement. Through the automation of critical processes and the integration of feedback loops, CI/CD empowers organizations to furnish top-tier, feature-enriched network infrastructure that aligns with the evolving requisites of contemporary telecommunications environments.
3) **xApps, rApps and dApps:** Within the framework of O-RAN, xApps, rApps, and dApps denote distinct varieties of applications or software components intended to augment and broaden the functionalities of the open and disaggregated RAN architecture. These categories of applications or software components - xApps, rApps, and dApps - contribute to the openness, flexibility, and innovation of the O-RAN architecture. They empower operators to expand the capabilities of the RAN infrastructure, integrate specialized functionalities, and harness cloud-native architectures to provide scalable and agile network services.
4) **3GPP and non-3GPP co-existence:** In O-RAN deployments, the coexistence of 3GPP and non-3GPP technologies pertains to the integration and interoperability of conventional cellular network technologies standardized by the 3GPP with emerging or non-standardized wireless technologies within the O-RAN architecture. This coexistence enables operators to harness a diverse array of wireless technologies, spectrum bands, and deployment options within O-RAN deployments, catering to evolving connectivity requirements and offering distinctive services to end-users. Embracing principles of openness, interoperability, and flexibility, O-RAN architectures empower operators to unlock the full potential of heterogeneous network environments and spur innovation in the telecommunications sector.
5) **Quantum for O-RAN:** QC holds promise for transforming numerous facets of telecommunications, including O-RAN deployments. Although still in its nascent stages, QC presents various potential areas of impact on O-RAN. It's crucial to acknowledge that quantum computing remains an emerging field, and practical applications in telecommunications, such as O-RAN, are still undergoing exploration. While the potential advantages of quantum computing for O-RAN are encouraging, substantial research and development endeavors are requisite to actualize these benefits and surmount technical obstacles related to quantum hardware, algorithms, and integration into current network infrastructure.

In this section, we introduced a comprehensive taxonomy for O-RAN in the context of 6G networks, categorizing the most recent contributions into four primary areas: architecture, technologies, challenges, and open research directions. This taxonomy serves as a foundational resource for understanding the design elements and evaluating proposed solutions aimed at addressing the key performance requirements of 6G, including EE, and LL. Through a systematic analysis, we identified how an O-RAN architecture facilitates disaggregation, virtualization, and the use of AI/ML for intelligent network management, enabling it to tackle emerging 6G use cases such as digital twins, V2X communications, and mMIMO. Additionally, we highlighted critical challenges, such as interoperability, integration complexity, and the need for enhanced security, which must be addressed to ensure seamless O-RAN deployment. Finally, we explored promising open research directions, including the integration of AI for security, CI/CD, the role of dApps, coexistence with traditional RAN, and the potential impact of quantum computing. By addressing these directions, future O-RAN implementations will be well-positioned to meet the demands of 6G networks, offering scalable, efficient, and secure solutions for next-generation wireless communication systems.

V. An O-RAN Architecture for EE

Building on the comprehensive taxonomy introduced in Section IV, which categorized the latest contributions into architectures, technologies, challenges, and open research directions, we now focus on the practical realization of energy-efficient O-RAN architectures. EE is a critical requirement for 6G networks, driven by the need to support massive connectivity, reduce operational costs, and address environmental concerns through sustainable solutions. O-RAN's disaggregated and open architecture provides a unique opportunity to optimize energy consumption across hardware and software components while leveraging AI-driven resource management and advanced hardware accelerators. In this section, we examine how various architectural frameworks and innovations in hardware, virtualization, and intelligent orchestration contribute to achieving energy-efficient O-RAN deployments. By analyzing the interplay between these elements, we highlight key design principles, technologies, and solutions that enable significant energy savings without compromising performance, scalability, or flexibility—essential factors for realizing 6G's ambitious energy targets.

An O-RAN architecture plays a significant role in advancing EE, which is crucial for future networks like 5G and 6G that need to handle increased data traffic, reduce carbon footprints, and support a massive number of devices. O-RAN aims to disaggregate traditional, monolithic RAN architectures by splitting hardware and software components, enabling multi-vendor environments, and leveraging cloud-native solutions. This disaggregation allows for more flexible resource management and use of energy-efficient hardware.

A. Key Features of O-RAN for EE

Virtualization and Cloud-Native Architecture: O-RAN enables the use of cloud-native architectures where network functions can be virtualized and placed on COTS hardware. This flexibility allows operators to dynamically scale network resources based on demand, which reduces energy consumption during low traffic periods [23].

AI and Machine Learning for Resource Optimization: O-RAN's integration of AI and ML is critical for optimizing resource allocation dynamically. AI can predict traffic patterns and adjust the usage of RUs, DUs, and CUs to optimize energy consumption. For example, AI-driven energy-saving techniques can put certain network elements into sleep mode or reduce transmission power when traffic is low, leading to significant reductions in energy use [24].

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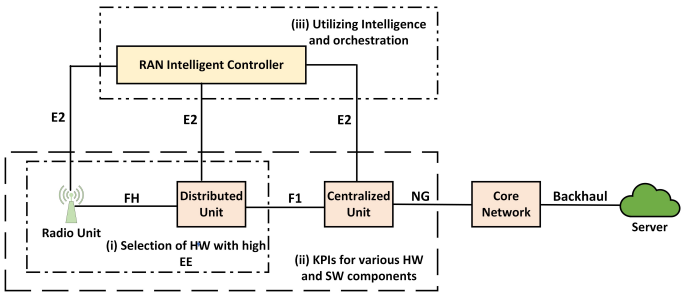


Figure 10. Foundational elements for EE in O-RAN.

Dynamic Resource Scaling: O-RAN's disaggregation allows for the independent scaling of various network components, such as the RUs, DUs, and CUs. This capability ensures that energy is only consumed based on the actual network demand, unlike traditional RAN where base stations consume a fixed amount of energy irrespective of the traffic load. Example: During periods of low traffic, the O-RAN components can be reconfigured or powered down, and dynamic spectrum allocation can be employed to reduce energy consumption in specific frequency bands [25].

Energy-Efficient Hardware: O-RAN enables the use of energy-efficient hardware components, such as low-power radio units and accelerators like FPGAs or ASICs for specific tasks like encoding/decoding or beamforming in massive MIMO systems. These specialized hardware units are designed to be more energy-efficient than general-purpose processors. Power Amplifier Optimization: O-RAN allows for energy-efficient power amplifiers in RUs that can switch to low-power states during idle periods.

EE plays a pivotal role in advancing mobile networks, particularly in the context of 5G and beyond. Within the RAN, it has become a focal point for both industry and research endeavors. With the ongoing push for higher throughput's in these advanced networks, there is a growing demand for energy-intensive hardware deployment. Consequently, the implementation of energy-saving mechanisms becomes imperative. The O-RAN Alliance is actively engaged in facilitating discussions and initiatives aimed at supporting energy-saving initiatives, recognizing it as a fundamental pillar for the progress of the RAN domain [23].

EE is a comprehensive necessity that spans across all domains of O-RAN, and it is advisable to prioritize its consideration in the initial stages of designing O-RAN solutions. The overarching goal is for O-RAN networks to evolve to be increasingly energy efficient compared to traditional RAN setups, all while maintaining key O-RAN principles like cloudification and disaggregation [24]. EE for O-RAN networks should thus be grounded on the following foundational elements:

1) Selection of HW with high EE.
2) Key Performance Indicators (KPIs) for various hardware and software components to assess EE.
3) Utilizing intelligence and orchestration for automating EE functionalities.

B. Selection of HW with High EE

Utilizing energy-efficient hardware is crucial for minimizing network energy consumption. Although O-RUs constitute a significant portion of power usage in the RAN, it is important to also account for the hardware infrastructure supporting other functions like O-DUs. A starting point involves designing Power Amplifiers not only with energy-efficient power amplification and transceivers but also with the capability to deactivate most transmitter elements and digital front ends when the network experiences low loads. COTS servers that rely on General Purpose Processors (GPP) for DU functions should ideally incorporate power-efficient hardware, including modules with low voltage requirements. The selection of accelerator technology (such as field programmable gate arrays (FPGAs), graphics processing units (GPUs), or application-specific integrated circuits (ASICs)) for supporting compute-intensive tasks like mMIMO should be made carefully, considering their power efficiency. Globally, to ensure optimal EE performance, recommended targets have been established for each hardware element within the network.

The computational demands of 5G RAN have notably increased in comparison to previous generations. This is due to essential features such as wider channel bandwidths, facilitation of LL, implementation of intricate channel coding schemes like low-density parity check (LDPC) code and polar code, as well as the integration of advanced antenna technologies featuring mMIMO antenna arrays [25]. Hence, relying solely on GPPs to implement 5G RAN can result in unacceptable levels of power consumption, rendering network deployment impractical. To address this issue, it is essential to harness the capabilities of hardware acceleration, diverting compute-intensive RAN tasks to specialized hardware devices such as GPUs, data processing units (DPU), ASICs, FPGAs, digital signal processors, or system-on-chips(SoC). This approach enables accelerated computation for 5G RAN, mitigating power consumption concerns and enhancing overall efficiency [17]. The significance of hardware acceleration for O-RAN is gaining increasing recognition within the wireless and networking communities.

*1) **Lookaside and Inline HW Acceleration**:* In the RAN protocol stack, the workload of gNB-DU processing scales with factors such as the number of connections, transmission time interval (TTI), bandwidth, and antenna branches. Typically, the lower the layer in the protocol stack, the greater the computational demand. Specifically, the demand for PHY layer processing within gNB-DU grows exponentially in tandem with the bandwidth and number of antenna branches. For instance, in the case of a 5G mid-band carrier featuring 30 kHz subcarrier spacing, 100 MHz bandwidth, and 64 transmit and receive antennas, approximately 85% of the processing requirements of the RAN protocol layers are attributed to the PHY layer, including tasks such as beamforming [25].

The increasing demand for L1 processing in 5G RAN requires accelerated computing, and often this cannot be met by GPPs alone. Relying solely on central processing unit (CPU)-based platforms for 5G RAN can lead to impractical levels of cost and power consumption. Hence, hardware acceleration becomes indispensable, enabling computationally intensive PHY layer signal processing tasks to be offloaded from the host CPU to specialized hardware accelerators (HWAs). HWAs are specialized hardware implementations designed to handle compute-heavy workloads more efficiently than GPPs. The selection of HWAs varies, ranging from fully programmable processors such as GPUs and DPUs to heavily customized, mostly fixed-function accelerators like FPGAs, DSPs, ASICs, and SoCs. Through the utilization of HWA, the PHY layer can be implemented either as a purely hardware solution, running entirely on the HWA, or as a hybrid 'software plus hardware' solution, where processing is distributed between the CPU and HWA depending on the specific use case and the level of accelerated computation required [17].

Typically, HW acceleration can operate in either lookaside or inline mode [26], as depicted in Fig. 11. In lookaside mode, the host CPU initiates data processing offload to a HWA and subsequently receives the processed data back from the HWA once the computation is finished. This mode has been predominantly utilized for traditional RAN accelerations for the PHY layer since the 4G LTE era, such as forward error correction (FEC) for physical control and shared channels. Channel coding stands out as one of the most complex signal processing tasks within Layer 1. An FEC accelerator facilitates faster and more efficient processing of various channel encoding/decoding algorithms compared to relying solely on the CPU [27]. A drawback of lookaside acceleration is the back-and-forth data transfer between the host CPU and the HWA. When multiple non-contiguous functions are accelerated in lookaside mode, as illustrated in Fig. 11, the overhead of repetitive host-to-device data transfer increases substantially, leading to higher double data rate bandwidth consumption. With the emergence of mMIMO antenna

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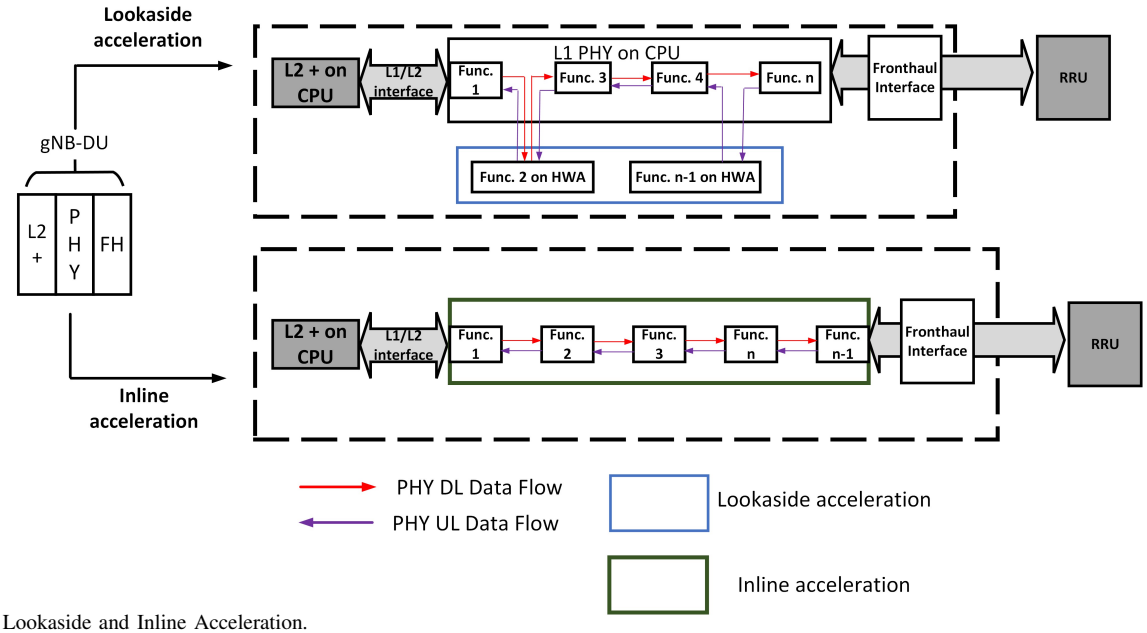


Figure 11. Lookaside and Inline Acceleration.

arrays, coupled with the deployment of new and broader 5G mid/high frequency bands and shorter TTIs, the computational demands for 5G Layer 1 processing are growing exponentially. This growth is pushing the limitations of lookaside acceleration to the edge. Consequently, inline acceleration mode has emerged as a promising solution for alleviating the bottleneck caused by massive data transfers in the lookaside approach [27].

The fundamental principle of inline PHY acceleration involves offloading the entire L1 processing pipeline to the HWA. Upon completion of processing, the output data is sent directly from the HWA to the next destination (such as the Remote Radio Unit or RRU), bypassing the CPU. This approach circumvents the redundant data path between the CPU and HWA, which often throttles data transfer rates. Facilitating fast input/output processing, for example, through the utilization of technologies like Smart Network Interface Cards SmartNICs and Direct Memory Access [28], is crucial in realizing the full potential of inline acceleration.

In [27], the authors provide insights into the implementation specifics of an FPGA accelerator design and the integration process of this accelerator with OpenAirInterface, serving as the foundation for their 5G NR platform. They demonstrate a fully integrated FPGA-based LDPC accelerator within a comprehensive software platform, capable of achieving decoding throughput exceeding 1.6 Gbps and exhibiting execution times up to 13 times faster than single-core software implementations. The thorough investigation conducted in [29] offers a holistic comprehension of the latest advancements in ASIC, FPGA, and GPU-based systems. It sheds light on the prevailing challenges that need to be addressed for the development of more efficient Non Binary-LDPC decoder architectures. While this study may not be directly aligned with the focus of this survey, it provides a broader perspective on the efficiency advantages of Non Binary-LDPC decoding compared to traditional LDPC decoding, as well as insights into the hardware platforms suitable for executing NB-LDPC decoding tasks.

C. KPIs for Various HW and SW Components

Measuring and reporting energy consumption is a fundamental necessity for effectively managing EE throughout the network. As such, each network element will necessitate specific KPIs. The subsequent counters and KPIs have been given priority:

1) The availability of counters and KPIs enables continuous monitoring of power consumption per Radio Access Technology (RAT) and frequency band/cell in Multi-band O-RUs, regardless of the time period.
2) O-RU KPIs encompass EE and power consumption metrics derived from real-time metering.
3) KPIs for O-CU/O-DU hardware and software, as well as O-Cloud software platform, are advised to be:
   a) O-CU/O-DU hardware components such as CPU, accelerators, NIC cards, fans, and power supply units are required to possess the ability to measure and report power consumption values to the O-Cloud.
   b) The O-Cloud software must be capable of gathering measurement data at the hardware component level (e.g., CPU, NIC, accelerator card, fans, and power supply units) and offer parameters and KPIs related to power, energy, and environmental aspects. Additionally, the O-Cloud should be able to report EE either through the O2 interface to the Service Management and Orchestration (SMO) or through Northbound Interface (NBI) to external tools.
   c) The O-Cloud software should be capable of delivering Power, Energy, and Environmental parameters and KPIs at the workload level (e.g., pod, O-CU/O-DU CNF, etc.), as well as for the O-Cloud platform software components. Furthermore, the O-Cloud must have the capability to report EE through the O2 interface to the SMO or through NBI to external tooling.
   d) O-CU and O-DU should offer Container Network Function level EE counters and KPIs, such as power consumption, traffic load, data volume, and throughput. These metrics should be reported through the O1 interface to the SMO or through NBI to external tooling.
4) It is anticipated that all KPIs will necessitate customized collection and reporting mechanisms:
   a) Reporting KPIs should be facilitated across O-RAN interfaces.
   b) Data collection and dashboard generation should occur at the SMO level, allowing for various levels of aggregation, including cluster, CNF, and PNF levels.
   c) A hierarchical or hybrid management model for the O-RU should produce comparable KPI results.

Additionally, to measure the power consumption of RAN network elements such as O-RU, O-CU/O-DU software and hardware, and O-Cloud software platform under realistic load scenarios, RF load generators will be essential. These generators should have the capability

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Table IV
COMPARISON OF O-RAN SOLUTIONS FOR EE

Approach	Key features	Advantages	Challenges	Examples from Literature
Dynamic Resource Scaling	Dyanmic adjustment of RAN Components based on traffic demand	Significant power savings during low traffic	Requires precision AI/ML integration	Described in [Abubakar et al., [16]]
Energy-Efficient HW	Use of low-power RUs and accelerators (ASICs, FPGAs)	HW optimized for specific tasks like mMIMO processing	Intitial deployment costs, integration into legacy systems.	Highlighted in [Azariah et al., [9]]
AI-driven Resource Optimization	Traffic prediciton and intelligent management of DU/CU functions	Reduces wastage, enables sleep modes	High Computational cost for AI Algos	Discussed by O-RAN Alliance and [13]
Cloud Native Platform	Virtualization and containerization of DU/CU	Flexible scaling, dynamic resource allocation	Requires optimized orchestration and control	[Polese et al., [13]]
Power Amplifier Optimization	Low-power amplifiers with adaptive states	Reduces O-RU energy consumption	Complex implementation for multi-vendor RAN	[Abubakar et al., [16]]

to artificially load the network by activating pre-defined numbers of Physical Resource Blocks.

The authors in [30] conducted experimental research focusing on measuring throughput, delay, and delay jitter as KPIs for achieving high-quality holographic telepresence applications. Their study is based on a wireless RAN utilizing an O-RAN architecture. The hardware load, round trip time, throughput, and jitter measurements and analysis are presented, along with the results of the end-to-end speed test.

D. Key Features and Comparisons of Energy-Efficient O-RAN Solutions

The O-RAN architecture facilitates EE through virtualization, AI-driven resource management, and hardware innovations. To provide a clearer understanding of the trade-offs and benefits, Table.V, Table.VI and Table.VII present detailed comparisons of various approaches proposed.

E. Conceptual Evaluation of O-RAN Strategies

To further elaborate on the advantages and disadvantages of these approaches, Table.VIII provides a comparative summary of the diverse proposed solutions.

F. Recommendations and Future Directions

Hybrid Approaches: Combine virtualization and hardware acceleration to optimize both cost and efficiency. For example, leveraging ASICs for critical tasks within a virtualized framework can reduce latency while maintaining flexibility.

Dynamic Split Selection: Implement AI-based systems to dynamically switch between split options (e.g., Split 6 or Split 7-2x) based on traffic patterns and deployment scenarios.

Energy Efficiency KPIs: Standardize metrics for energy consumption across O-RAN components, enabling consistent evaluation and optimization.

Integration with Emerging Technologies: Explore synergies with quantum computing and digital twin technologies to enhance O-RAN's capabilities in complex scenarios.

In conclusion, the adoption of O-RAN presents a substantial opportunity to improve EE within telecommunications networks. Through the breakdown of network functions, fine-tuning resource distribution, and harnessing cutting-edge technologies such as AI and machine learning, operators stand to lower energy usage, decrease operational expenses, and play a role in fostering a greener trajectory for the telecommunications sector. Nonetheless, sustained research, cooperation, and standardization endeavors remain imperative to unlock the complete potential of O-RAN in advancing energy-conscious network deployments.

VI. O-RAN ARCHITECTURE FOR LL

As highlighted in Section IV, the evolution of O-RAN architectures and technologies presents significant opportunities to address the diverse requirements of 6G networks, including energy efficiency, low latency, and high throughput. While energy efficiency was explored in Section V, the focus now shifts to LL critical pillars for enabling advanced 6G use cases such as real-time communications, autonomous systems, and immersive applications. O-RAN's flexible, open, and disaggregated design allows for the deployment of innovative solutions, including edge computing, dynamic resource allocation, and AI-driven optimization, to meet these stringent performance demands. In this section, we analyze various approaches and technologies proposed in the literature to reduce latency in O-RAN deployments. By comparing architectural strategies, intelligent management techniques, and hardware optimizations, we aim to showcase how O-RAN can deliver the URLLC and high-capacity performance required for 6G applications. This discussion further emphasizes O-RAN's role as a key enabler for 6G, providing solutions that align with its ambitious goals of delivering real-time, high-speed, and scalable wireless communication systems.

LL O-RAN is a key approach in telecom aimed at reducing communication delays within cellular networks, particularly for applications that demand ultra-fast responses, like real-time gaming, AR/VR, and critical IoT functions in smart cities or autonomous vehicles. Traditional RANs are often proprietary and hardware-focused, while O-RAN introduces more flexibility and cost savings by decoupling hardware and software components, fostering an ecosystem where components from different vendors can interoperate. Benefits of LL O-RAN Enhanced User Experience: LL services improve real-time applications, making them more responsive and reliable. Support for Emerging Applications: Technologies like autonomous driving, industrial IoT, and telemedicine benefit from O-RAN's low latency, which can facilitate faster data exchange. Lower Costs and Improved Flexibility: O-RAN, with its focus on software-based solutions and open standards, offers potential cost savings and allows operators to adapt their networks for different use cases without large investments in proprietary hardware.

The O-RAN architecture can be optimized for LL by implementing several key strategies:

1) **Edge Computing Integration:** Latency may be greatly decreased by placing edge computing resources closer to end users at the network edge. By integrating edge computing nodes, open RAN design minimizes latency by allowing processing operations to be carried out closer to the source of data.
2) **CU and DU Split:**The CU, which handles higher-layer processing, and the DU, which handles lower-layer tasks, are divided in an O-RAN design. This division enables the CU to be centralized for effective resource use, while the DUs can be

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Table V
KEY FEATURES AND COMPARISONS OF ENERGY-EFFICIENT O-RAN SOLUTIONS

Feature	Approach	Advantges	Disadvantges	Use cases
Virtualization	Cloud-Native RAN	Flexibility, dynamic scaling, cost reduction through COTS HW	Potential latency and energy overhead	Scalable urban networks
AI/ML for Resource Optimization	Predictive traffic models	Dynamic scaling, energy savings through sleep modes	Complexity in implementation, reliance on accurate data	Smart cities, autonomous vehicle management
HW Acceleration	ASICs, FPGAs, GPUs	High Efficiency in processing-intensive tasks (e.g., mMIMO, LDPC decoding)	High Initial cost, limited flexibiltiy	Dense urban areas with high data throughput
Split Options (7-2x vs Split 6)	Split 7-2x	Supports mMIMO. better suited for urban macrocells	Increased FH bandwidth requirements	High-capacity urban deployments
	Split 6	Lower FH BW, suitable for small cells and rural areas.	Limited supports for advanced RF techniques like mMIMO.	Rural deployments with lower traffic density.

Table VI
VIRTUALIZATION VS. HARDWARE ACCELERATION

Parameter	Virtualization	HW Acceleration
EE	Moderate; depends on intelligent scaling	High; optimized for specific tasks
Scalablilty	Highly scalable; leverages cloud infrastructure	Limited scalability; HW updates required
Latency	Potential for increased latency due to virtualization	Ultra-low latency in processing-intensive operations
Cost	Lower initial cost but higher operational costs	High initial cost, lower long-term operational costs
Flexibility	Highly flexible; supports multi-vendor environments.	Limited flexibility

Table VII
SPLIT 7-2X VS. SPLIT 6

Aspect	Split 7-2x	Split 6
Flexibility	Suitable for advanced techniques like mMIMO	Limited flexibility
Bandwidth Requirements	High due to frequency-domain IQ data transmission	Lower, using transport layer simplifications
Energy Consumption	Increased due to additional processing requirements	Lower power consumption
Use Case	Dense urban macrocells	Rural and small-cell deployments

Table VIII
CONCEPTUAL EVALUATION OF O-RAN STRATEGIES

Approach	Advantages	Disadvantages
Virtualization	Cost-effective, flexible, enables multi-vendor solutions	May introduce latency and energy inefficiency in high-demand scenarios
AI/ML Optimization	Enhances real-time decision-making, improves energy efficiency	Dependency on accurate data, risk of AI model inefficiencies
Hardware Acceleration	Provides ultra-low latency, optimized for computationally demanding tasks	Higher cost and reduced adaptability
Split 7-2x	Supports advanced RF processing, aligns with urban high-throughput requirements	Higher FH bandwidth and power requirements
Split 6	Simplified design for lower-demand areas, cost-efficient for small cells	Cannot support advanced RF functionalities like mMIMO

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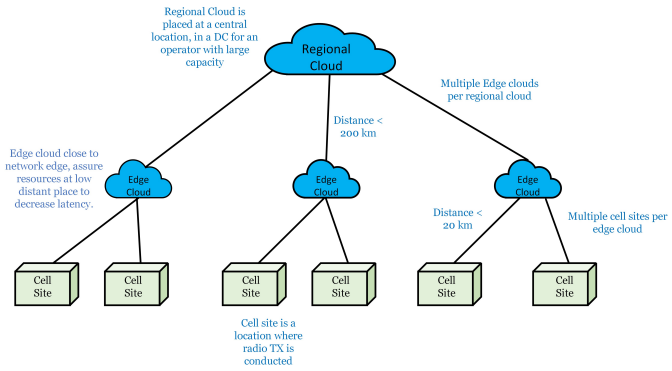


Figure 12. O-RAN Deployment Scenarios.

deployed dispersed, bringing them closer to the end-user for lower latency.

3) **Functional Decomposition:** O-RAN divides network operations into more manageable, modular parts. Because of this breakdown, certain functions may be deployed where they are most needed, minimizing the distance over which data must travel and, as a result, latency.
4) **Real-time Orchestration and Optimization:** Real-time dynamic orchestration and network resource optimization are supported by O-RAN architecture. By optimizing latency-sensitive traffic pathways, this technology enables intelligent routing decisions and resource allocation depending on current network circumstances.
5) **Network Slicing:** Network slices that are customized to meet certain latency needs may be created thanks to open RAN. Effective support for low-latency services, including ultra-reliable low-latency communication (URLLC), can be achieved by assigning certain resources and giving traffic inside these slices priority.
6) **Interoperability with Low-Latency Technologies:** Emerging low-latency technologies like Time-Sensitive Networking, Multi-Access Edge Computing (MEC), and 5G URLLC can be integrated with O-RAN architecture. This integration guarantees smooth interoperability and improves the network's overall latency performance.

Operators can create and implement networks optimized for LL, satisfying the needs of latency-sensitive applications and services in a variety of use cases, such as gaming, virtual reality, autonomous vehicles, and industrial automation, by incorporating these components into O-RAN architecture design and implementation.

A. O-RAN Deployment Scenarios

The deployment scenarios is define by O-RAN alliance to define how to compose the elements of O-RAN architecture (CU, DU, RU, Near-RT RIC) in the actual network deployment within the cloud and physical locations (as shown in Fig. 12). It is defined by WG6. Split 7 is the major aspect of the deployment perspective, because it gives you Physical Network Functions (PNF) (O-RU) deployed as HW and Virtual Network Functions deployed at O-Cloud (CU, DU). O-Cloud is a cloud computing platform comprising physical infrastructure nodes to host O-RAN functions, like Near-RT RIC, O-DU, etc.; supporting software components (e.g., Operating System, Virtual Machine monitoring, container runtime), management, and orchestration functions. For eMBB-mMTC services, are latency constraint applications, the deployment may be more centralized, where O-CU deployed in the regional and O-DU in edge (as shown in Table. IX). For URLLC services, the O-CU shall be deployed closer to the user, O-CU and O-DU at the edge cloud.

B. Six Scenarios

Scenario A: The whole base station is deployed at the edges as VNFs in the O-Cloud, while O-RUs are distributed to cell sites as PNF. This minimizes the delay, but the cost of such deployment can be the largest compared to other scenarios (as shown in Table. X).

Scenario B: In scenario B, the actual processing of the base station (i.e., O-CU and O-DU) is deployed at the edge, so we keep LL, but the Near-RT RIC is put centrally to the regional cloud DC to capture a broader view of the network and this is the initial use case adopted by O-RAN (as shown in Table. X).

Scenario C: In scenario C, only the O-DU is at the edge, while O-CU to the regional DC to save cost and decrease energy consumption at the edge location. O-CU, in general, is working on a similar time scale as Near-RT RIC and thus they can be put together. Only O-FH has stringent latency requirements, while E2/F1 interfaces can be more relaxed (as shown in Table. X).

Scenario D: In scenario D, the setup is as in scenario C, but the O-DU is realized as a physical network function/device. This option enables the use of high-performing and optimized hardware for MAC/high-PHY layer processing (as shown in Table. X).

Scenario E: In scenario E, everything is virtualized – down to O-RU (which, of course, can use hardware accelerators). In such a case, put O-DU together with O-RU at the cell site with the high-performing COTS hardware for PHY layer processing. This is considered for future use as the virtualized versions of the low-PHY layer and other O-RU aspects are not yet there (as shown in Table. X).

Scenario F: Scenario F, all elements are also virtualized (as in scenario E), but with different locations for O-RU and O-DU. This scenario is also (as scenario E) considered for future use for a similar reason (as shown in Table. X).

In [31], a framework based on O-RAN is introduced for predictive UL slicing (shown in Fig. 14). It presents an xApp that utilizes deep learning to dynamically adjust RAN scheduling via the RIC's E2 interface. The framework's evaluation is conducted using a demanding scenario involving URLLC traffic from the Smart Grid domain, implemented in an experimental laboratory environment. Despite the introduction of additional interfaces by O-RAN, the framework achieves performance comparable to proprietary solutions, with latencies reaching as low as 5 ms.

[32] outlines the proposal and implementation of a low-latency Near-RT RIC platform along with an autonomous intelligent xAPP generation design. The goal appears to be the adaptation of universal vertical applications efficiently within the context of the O-RAN Alliance's control loop requirements. The authors intend to demonstrate that their proposed Near-RT RIC platform effectively meets the control loop requirements set by the O-RAN Alliance. They evaluate the performance of the platform using vertical applications such as conversational voice, video/gaming streaming, and V2X. The testing results validate the effectiveness of their autonomous intelligent xAPP generation design in supporting these 5G vertical applications.

In [33], authors describes a discussion on the use of hardware-enforced capabilities as a foundational concept for future O-RAN architectures. The authors introduce their strategy of a hardware/operating system co-design to implement these principles effectively (as shown in Fig. 15). Through evaluation, they present results that demonstrate the technical feasibility of their approach. Specifically, they illustrate that their architectural components can achieve both security and LL simultaneously. This suggests a promising direction for designing O-RAN architectures that are robust, secure, and efficient.

In [34], the authors tackle the crucial requirement for effective multi-time-scale control loops within open RAN deployments, particularly for supporting uRLLC services. They introduce ORANUS, a mathematical framework compliant with O-RAN standards, focusing on radio resource allocation problems at both near-RT and RT scales. For the near-RT control loop, ORANUS employs a novel Stochastic Network Calculus (SNC) model. Unlike traditional queueing theory approaches, the SNC model enables ORANUS to ensure that the probability of packet transmission delay exceeding a specific budget (violation probability) remains below a target tolerance. Additionally, ORANUS incorporates an RT control loop, which monitors the transmission queue of each service and dynamically adjusts the

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Table IX
DIFFERENT LOCATIONS

Where to put?					Advantages	Disadvantages?
RU	**DU**	**CU-CP**	**CU-UP**	**Near-RT RIC**		
Cell Site	Cell Site	Cell Site	Cell Site	Cell Site	Lowest possible latency for processing.	Deployment is the most expensive one, cell site requires all functionality.
Cell Site	Cell Site	Cell Site	Edge Cloud	Edge Cloud	Collected Manage the resources from the connected cell site	Process the data including the user plane traffic at cell sites, but reduce latency
Cell Site	Edge Cloud	Edge Cloud	Regional Cloud	Regional Cloud	Implements a cheaper deployment with less processing power needed at each cell site. Putting CU, nE-RT RIC at regional cloud, gives freedom of gathering processing from more cell sites.	Increases latency for data processing

Table X
SIX POSSIBLE DEPLOYMENT SCENARIOS

Scenarios	O-RU	O-DU	O-CU-CP	O-CU-UP	Near-RT RIC
Scenario A	PNF	Edge cloud	Edge Cloud	Edge Cloud	Edge Cloud
Scenario B	PNF	Edge cloud	Edge Cloud	Edge Cloud	Regional cloud
Scenario C	PNF	Edge Cloud	Regional Cloud	Regional Cloud	Regional Cloud
Scenario D	PNF	PNF	Regional Cloud	Regional Cloud	Regional Cloud
Scenario E	Regional Cloud	Regional Cloud	Regional Cloud	Regional Cloud	Regional Cloud
Scenario F	Cell Site (VNF)	Regional Cloud	Regional Cloud	Regional Cloud	Regional Cloud

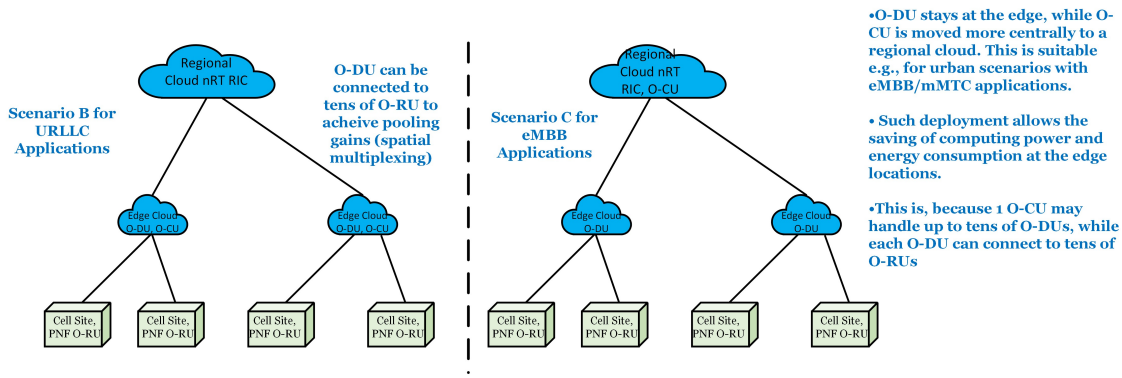


Figure 13. Deployment Scenarios: Scenario B and Scenario C

allocation of guaranteed Resource Blocks in response to traffic anomalies. The authors evaluate ORANUS through a comprehensive simulation campaign. Results demonstrate significant improvements compared to reference solutions, with an average violation probability that is 10 times lower. This underscores the efficacy of ORANUS in enhancing the reliability and performance of uRLLC services within O-RAN-based deployments.

In [35], the authors introduce a new framework for orchestrating network intelligence within O-RAN systems. OrchestRAN is built upon O-RAN specifications and utilizes RIC xApps and rApps, as well as O-RAN open interfaces, to provide Network Operators with an automated orchestration tool (as shown in Fig. 16). This tool is designed to deploy data-driven inference and control solutions with various timing requirements. OrchestRAN incorporates orchestration algorithms with different optimality and complexity trade-offs to support applications with non-RT, near-RT, and RT requirements. The authors evaluated the performance of OrchestRAN and presented an O-RAN-compliant prototype by implementing a cellular network with 7 base stations and 42 UEs on the Colosseum network emulator. Experimental results showcased that OrchestRAN facilitates the seamless instantiating of O-RAN applications across different network nodes and time scales. Additionally, it was observed that OrchestRAN reduces message overhead over the O-RAN E2 interface by up to 2.6 times when deploying intelligence at the network edge.

In the paper [36], authors propose a reinforcement learning-based resource allocation scheme for the RIC in O-RAN, aiming to achieve ultra-low latency in 6G networks. Their method employs Double DQN to allocate both computing and network resources efficiently. Additionally, they devise an updating mechanism for the AI models within the Near-RT RIC, enabling it to process more requests promptly. Simulation results presented in the paper demonstrate the effectiveness of the proposed Double DQN scheme, particularly in terms of reward and the number of accepted requests. Notably, when incorporating cache, the Double DQN scheme outperforms the greedy algorithm significantly, accepting 2785 requests compared to 2160 requests, thus representing a 28% improvement.

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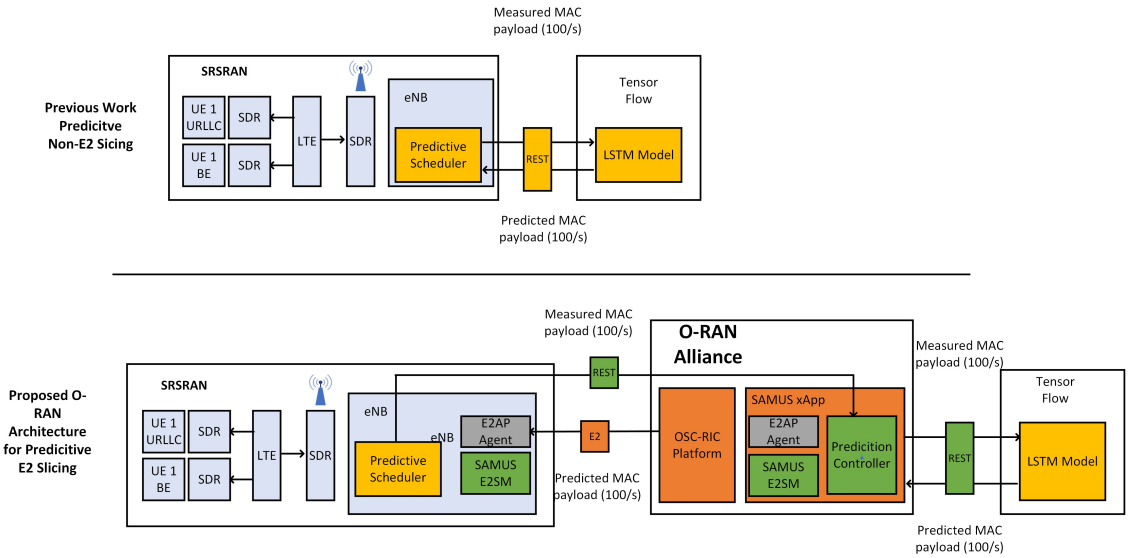


Figure 14. Depiction of the proposed O-RAN compliant framework for ML-driven UL slicing and its entirely open source-based elements and interfaces in contrast to a more traditional non-O-RAN setup presented in previous work [31].

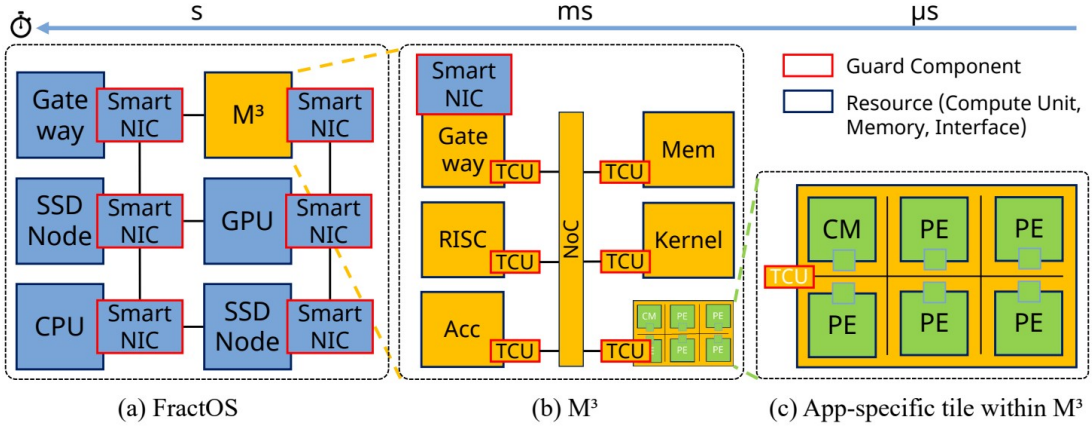


Figure 15. Approach on a secure hardware/OS platform. Multiple resources are organized hierarchically. Guard components enforce security policies and gateways enable communication across layers. FractOS may enclose one or more M3 systems, which in turn may enclose application-specific tiles with a CoreManager for low-latency data processing. [33].

The section discussing the low-latency architecture based on O-RAN outlines several innovative approaches. These approaches leverage the principles of O-RAN, employing techniques such as reinforcement learning-based resource allocation, orchestration frameworks, and hardware-enforced capabilities.

In one approach, a reinforcement learning-based resource allocation scheme is proposed for RIC deployment within O-RAN. This scheme utilizes Double DQN to efficiently allocate computing and network resources. Additionally, an updating mechanism for AI models in the Near-RT RIC enables quicker processing of requests, contributing to reduced latency.

Another approach is OrchestRAN, a novel network intelligence orchestration framework tailored for O-RAN systems. OrchestRAN leverages RIC xApps and rApps, as well as O-RAN open interfaces, to automate the deployment of data-driven inference and control solutions with diverse timing requirements. Experimental results demonstrate seamless instantiating of O-RAN applications across different network nodes and time scales, enhancing overall system performance.

Furthermore, there are discussions on hardware-enforced capabilities as a foundational concept for future O-RAN architectures. By co-designing hardware and operating systems, these capabilities aim to ensure security and LL simultaneously, thus providing a robust foundation for O-RAN deployments.

Overall, these approaches represent significant advancements in the quest for low-latency architectures within O-RAN, showcasing the potential to meet the 6G networks' stringent requirements, while enabling a wide range of applications with varying timing constraints.

C. Comparative Analysis of Key O-RAN Use Cases

The implementation of O-RAN has enabled numerous use cases, each requiring specific technologies and approaches. Below is a comparative summary of these solutions presented in Table.XII and Table.XIII, including notable use cases in the context of the solutions enabling them.

D. Conceptual Evaluations of Advanced O-RAN Technologies

Table.XIV summarizes the advantages and disadvantages of the major technologies critical to O-RAN use cases.

E. Recommendations and Future Directions

AI-Driven Optimization: Leverage advanced AI techniques to improve resource allocation, fault detection, and predictive analytics in Digital Twin and mMIMO use cases.

Hybrid Architectures: Combine centralized and distributed units to optimize latency and resource utilization for applications like V2X and URLLC.

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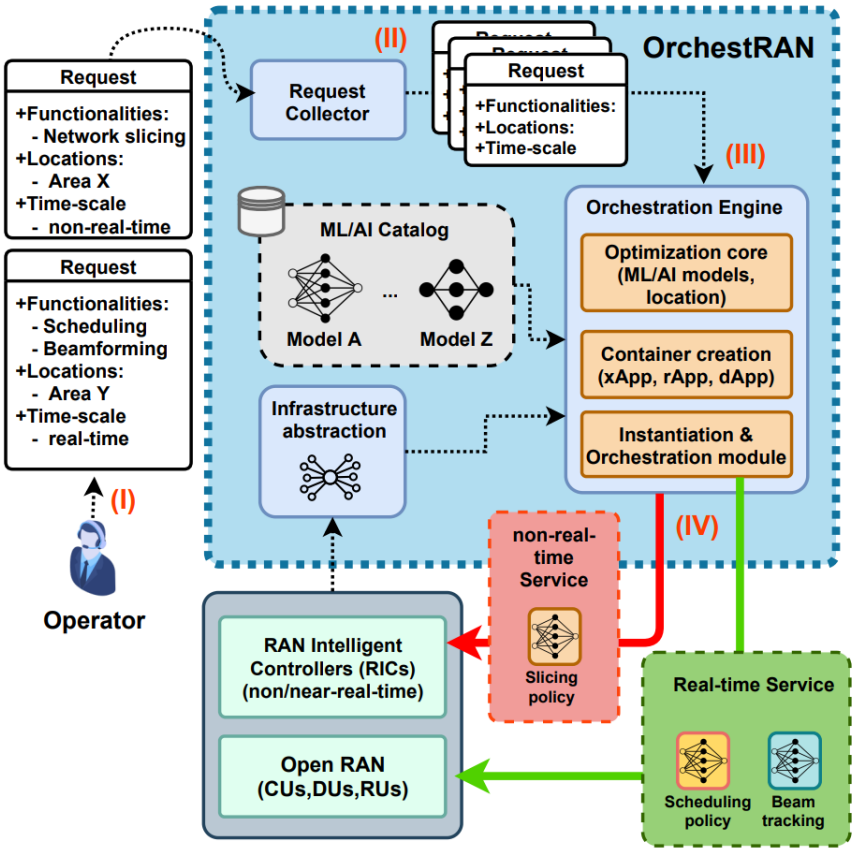


Figure 16. System design of OrchestRAN and main procedures [35].

Table XI
SOLUTIONS FOR LOW LATENCY (LL) OPTIMIZATION IN O-RAN

Approach	Key Features	Latency Reduction Benefits	Challenges	References
Edge Computing	Deploying processing functions at network edges	Minimizes round-trip delay	Requires edge infrastructure investment	[Polese et al., [13]]
Real-Time RIC (RT-RIC)	AI/ML-driven near-real-time optimization	Enhances real-time traffic steering	Computational overhead for RIC	[Wypiór et al., [12]]
Network Slicing	Allocates dedicated slices for latency-sensitive apps	Guarantees predictable latency performance	Complexity in resource orchestration	[Masaracchia et al., [17]]
Dynamic Spectrum Allocation	Frequency resources prioritized dynamically	Improves QoS for latency-critical services	Requires precise resource prediction models	[Polese et al., [18]]

Dynamic Slicing: Implement network slicing to dedicate resources dynamically for critical use cases such as V2X and anomaly detection.

Standardized Interfaces: Enhance the standardization of open interfaces to ensure interoperability without compromising security.

Energy-Efficient Protocols: Develop protocols specifically aimed at reducing power consumption in mMIMO and DT applications.

VII. INNOVATIVE TECHNOLOGY-RICH AVENUES IN O-RAN APPLICATION MIDDLEWARE CONTINUUM

In this section, we will delve into O-RAN application middleware continuum that utilizes the O-RAN architecture for their implementation. First, we will provide detailed discussions on technologies for supporting security, DT, V2X, and MIMO. Subsequently, we will review papers that have employed O-RAN in the context of these avenues.

A. Security

The security risks associated with the O-RAN architecture stem from its modular design and multi-vendor environment. Although modular design offers flexibility and scalability, it also gives rise to security concerns. Integrating components from various vendors may lead to compatibility issues and security vulnerabilities, increasing the system's susceptibility to attacks such as intrusion, identity fraud, and information theft [37].

1) **O-RU:** Malicious actors exploit technologies like SDR and Universal Software Radio Peripheral to execute harmful base station attacks against O-RUs [33]. This method leverages the extensive signal coverage to coerce nearby UE devices into connecting to the malicious base station [38]. Through this approach, attackers can acquire sensitive information such as the UE's Subscriber Privacy Identifier and 5G Global Temporary

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Table XII
COMPARATIVE ANALYSIS OF KEY O-RAN USE CASES

Use Case	Technology\ Approach	Advantges	Disadvantages	Key References
DT Integration	Real-time simulation and AI	Enables predictive analytics and network optimization	High computational requirements, potential latency issues	[Masaracchia et al., 2021], [DT Case Study]
mMIMO	Advanced RF beamforming	High spectral efficiency, essential for high-density urban deployments	High power consumption, complex hardware requirements	[Abubakar et al., 2021], [O-RAN Alliance Reports]
V2X Communication	AI-based traffic management	Supports ultra-reliable low-latency communication (URLLC)	High mobility poses synchronization challenges	[Polese et al., 2021], [Ali et al., 2020]
Energy-Efficient Networks	AI/ML optimization, dynamic RAN scaling	Significant energy savings through dynamic resource allocation	Dependence on accurate traffic forecasting, potential risks of under-provisioning	[Abubakar et al., 2020], [Green RAN Studies]
AI-Driven Anomaly Detection	Near-Real-Time RIC (nRT-RIC)	Enhances security by identifying threats dynamically	High reliance on robust AI models, data privacy concerns	[Polese et al., 2020], [Wypiór et al., 2021]

Table XIII
TRADE-OFFS IN IMPLEMENTING ADVANCED USE CASES

(A) DT Integration		
Aspect	**Advantages**	**Disadvantages**
Predictive Modeling	Real-time replication enables advanced analytics for optimization	High computational demands require significant resources
Fault Detection	Rapid fault identification reduces downtime	Synchronization challenges across distributed networks
Deployment Scalability	Applicable to smart cities and industrial automation	High costs for large-scale deployments
(B) mMIMO Deployment		
Spectral Efficiency	Optimized for dense urban environments	High power and hardware complexity
Coverage and Capacity	Improves coverage in challenging environments	Implementation costs are significant
Interference Mitigation	Beamforming reduces inter-cell interference	Requires advanced synchronization and calibration tools
(C) V2X Communication		
Safety	Enables real-time vehicular communication for collision avoidance	High mobility creates synchronization and handover challenges
Scalability	Supports large-scale connected vehicle ecosystems	Demands ultra-low latency, which may require additional resources
Cost	Improves traffic flow and reduces accidents	Cost of deploying roadside infrastructure

Identifier, ultimately resulting in the disconnection of the UE from legitimate base stations and causing service disruptions.

2) **Non-RT RIC:** Attackers can exploit vulnerabilities within the Non-RT RIC, gaining unauthorized access through the SMO interface, which presents several threats. Initially, attackers may track the location and behavior of UE, breaching user privacy. Additionally, they can manipulate information within the Non-RT RIC, leading to data corruption or manipulation and disrupting the system's regular operations. Furthermore, attackers may abuse these privileges for various malicious activities, including stealing sensitive data, launching Denial-of-Service attacks, or infiltrating the system, thereby posing severe security risks to the entire O-RAN network [39].
3) **Near-RT RIC:** The open platform of Near-RT RIC enables third-party xApp deployment, but it also introduces security risks. Malicious attackers can exploit this vulnerability by deploying malicious xApps, potentially compromising sensitive user data [40]. Users face a range of risks, including data theft, privacy breaches, financial losses, disrupted operations, service interruptions, compromised user experiences, and system-wide damage or intrusion [13].
4) **xApps:** The presence of vulnerabilities in xApps sourced from untrusted or poorly maintained origins can lead to severe consequences [13], [41]. Attackers exploiting these vulnerabilities can disrupt existing network services, infect other xApps, or compromise the entire Near-RT RIC, thereby impacting the O-DU and O-CU functionalities. Malicious xApps may excessively consume network resources, resulting in performance degradation using methods like Fork bombs. Additionally, attackers may launch adversarial attacks against AI/ML models within xApps, further disrupting the overall RAN operation [42], [43]. Moreover, these malicious xApps may gain unauthorized access to nE-RT RIC APIs or launch disruptive attacks such as DoS attacks. However, the current O-RAN specification only outlines potential threats without providing a detailed description of these threats and their occurrence processes. Furthermore, unknown threats may still persist, highlighting the ongoing need for robust security measures and threat intelligence in the O-RAN ecosystem [44]–[46].

In [47], the authors utilized a traffic steering module integrated with AD functionality. They conducted an evaluation and comparison of various ML algorithms for handover prediction within this module. The evaluated ML algorithms included local outlier factor, isolation forest, robust covariance, logistic regression, naive Bayes, support vector machine, and random forest. The AD xApp (as shown in Fig. 17) was responsible for identifying anomalous signal metrics affecting UEs within the network. This xApp retrieves anomalous UE data from the Influx database and undergoes ML training. Presently, within the O-RAN software, AD relies on the "isolation forest model" to identify anomalous UEs. The traffic steering xApp receives information about the type of anomaly or degradation detected, along with data related to the affected UEs.

In [49], the authors introduced an xApp designed to identify abnormal behavior exhibited by Industrial Internet of Things devices during their registration process. This xApp utilizes O-RAN interfaces to intercept Control Plane messages, allowing it to gather essential long-term network statistics necessary for detecting suspicious activity perpetrated by adversaries.

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Table XIV
CONCEPTUAL EVALUATIONS OF ADVANCED O-RAN TECHNOLOGIES

Technology	Advantages	Disadvantages
AI/ML in RAN Optimization	Dynamic allocation of resources reduces operational costs	Complexity in training models and reliance on high-quality data
Open Interfaces	Facilitates multi-vendor ecosystems and innovation	Security vulnerabilities due to increased attack surface
Virtualization	Reduces costs, scales dynamically	Potential latency overhead and increased operational complexity
Near-Real-Time RIC	Enhances decision-making with low latency for critical applications	Limited by the computational capacity and the accuracy of AI models
Distributed Units (DU)	Lowers latency by processing data closer to the edge	Higher deployment costs and infrastructure requirements
Beamforming for mMIMO	Improves spectral efficiency and throughput	Requires extensive calibration and power-intensive processing

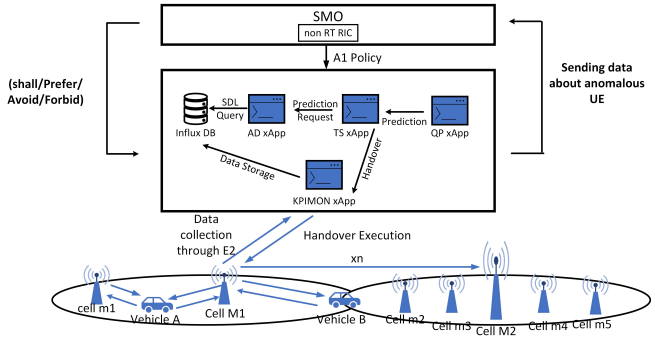


Figure 17. Traffic steering process for handover operations [48].

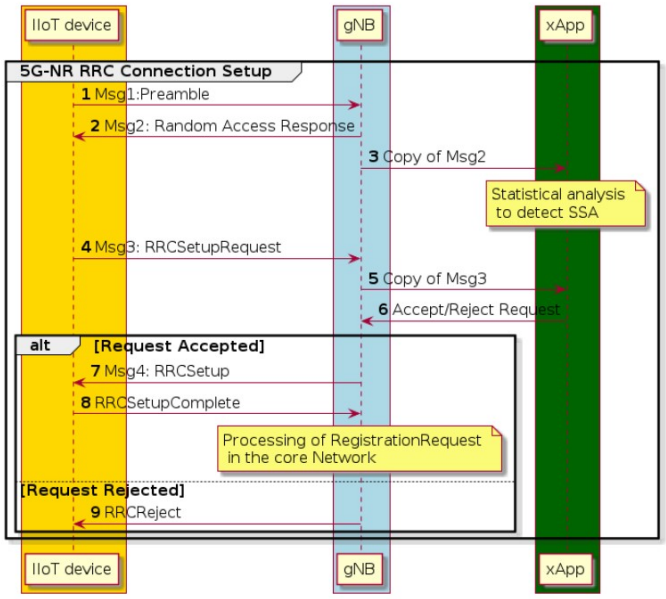


Figure 18. CP messages exchanged between IIoT device, gNB, and SSA detection xApp [49].

In [50], the authors suggested employing FL to detect anomalies within the O-RAN architecture while also safeguarding data privacy. They advocated for a peer-to-peer FL approach (as shown in Fig 19) to anomaly detection within the O-RAN framework and conducted an in-depth analysis of four different variants of P2P FL techniques. Additionally, they performed simulations of the proposed models using the UNSW-NB15 dataset.

In [51], a study is presented focusing on the feasibility and performance assessment of ML techniques, particularly supervised methods, for addressing anomaly detection challenges within a 5G communication network. The proposed ML models, including Multi-

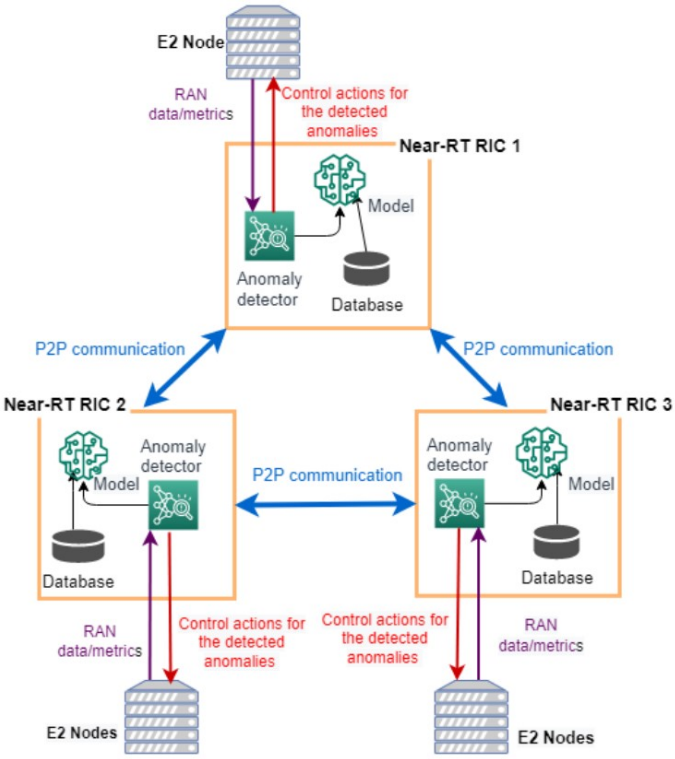


Figure 19. Proposed anomaly detection for O-RAN architecture [50].

layer Perceptron, Decision Tree, and Support Vector Machine, were employed to categorize data into either anomaly or non-anomaly categories, utilizing two distinct 5G O-RAN datasets featuring various KPIs. Additionally, they introduced a strategy aimed at identifying anomalous scenarios, leveraging the t-Distributed Stochastic Neighbor Embedding (tSNE) technique on datasets containing multiple KPIs. The findings demonstrated significant results, with accuracies exceeding 90% across all considered use cases. In [52], an approach for jamming detection (as shown in Fig. 20) was proposed. In this method, Channel Quality Indicator (CQI) and Reference Signal Received Power (RSRP) reports are gathered for a specific cell, considering a predetermined memory size that reflects the valid time horizon. This accounts for filtering out older reports collected under potentially different propagation conditions or levels of inter-site interference. If a report is not classified as collected under jamming, it is integrated into the current CQI-RSRP distribution estimate. Subsequently, a new report was compared with the current distribution, employing a local Gaussian approximation with a predetermined false alarm probability. If a given report deviates more significantly from the distribution than the specified false alarm threshold, it was inferred that jamming is occurring.

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Table XV
LATEST RESEARCH ABOUT ANOMALY DETECTION IN AN O-RAN ARCHITECTURE

Reference	How Mitigate Anamoly Detection?	At which entity?			
		O-RU	Fronthaul	CU/DU	RIC
[36]	The AD xApp is in charge of detecting anomalous signal metrics that serve UEs within the network.	x	x	x	In Near-RT RIC
[37]	A xApp to detect the abnormal activity of IIOT devices at the beginning of their registration procedure.	x	x	x	In Near-RT RIC
[38]	Federated Learning to detect anomalies.	x	x	x	In Near-RT RIC
[39]	Different ML models to detect anomalies	x	x	DU KPIs	x
[40]	A xApp for Jamming Detection	x	x	x	In RT-RIC

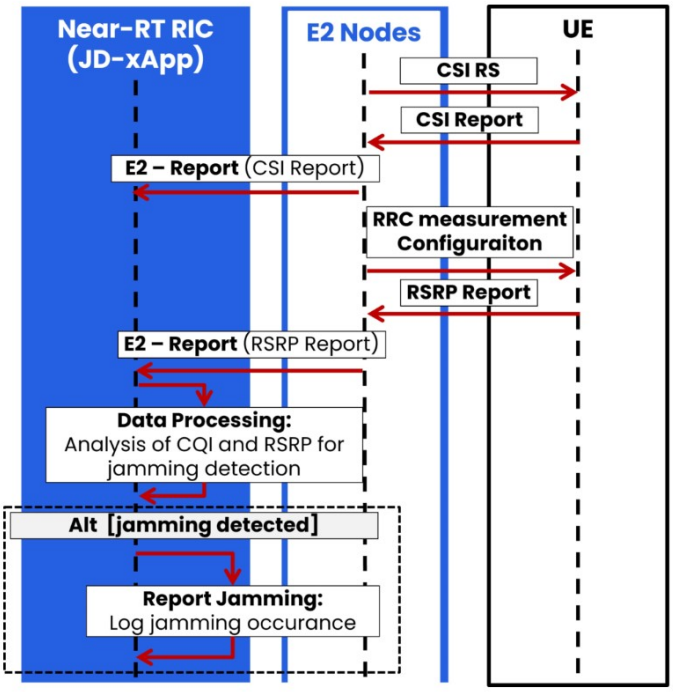


Figure 20. UML sequence diagram for jamming detection [52].

The principles of zero trust, authorization, authentication, and access control play a critical role in ensuring O-RAN security. Next is discussed how these principles apply.

1) **Zero Trust in the O-RAN Architecture:** The Zero Trust model assumes that no component, device, or user within the network is inherently trusted, even if they are inside the network perimeter [19], [91], [95]. In the context of O-RAN **Micro-Segmentation,** divides the network into micro-segments using its disaggregated architecture (e.g., CU, DU, and RU). Zero trust ensures that communication between these segments is controlled and monitored. **Continuous Verification** indicates that every interaction between O-RAN components (via open interfaces like E2, A1, and O1) is continuously verified. Even if a node was authenticated earlier, it must be re-verified based on behavior or access patterns. **Behavioral Analytics** is used to monitor network traffic to detect anomalies in communication between components (e.g., between a DU and CU), ensuring early detection of threats.
2) **Authentication in the O-RAN Architecture** Authentication ensures that only legitimate users and devices can access the network [19], [91], [95]. In O-RAN, there is a need for **Mutual Authentication,** as the open interfaces (e.g., O1 for management, A1 for policy control) require strong mutual authentication mechanisms between components. Both entities in a communication exchange (e.g., a DU connecting to a CU) validate each other's identities. There is also need for good **Device Identity Management, as** authentication is enforced through unique identities for each component (e.g., certificates for RU, DU, CU). This prevents impersonation attacks. Finally there is an interest to support **integration with Authentication, Authorization, and Accounting (AAA) Frameworks**, as O-RAN can leverage AAA protocols like RADIUS or Diameter to manage access to network resources.
3) **Authorization in the O-RAN Architecture:** Authorization ensures that even after authentication, users or devices can only access resources they are explicitly allowed to [19], [91], [95]. O-RAN implements this through the following avenues. **Role-Based Access Control (RBAC)** refers to the fact that components in O-RAN are assigned roles that dictate what operations they can perform. For example, a RU may only have limited access to spectrum resources, while a CU has more expansive permissions for data control. **Policy-Driven Access** relies on the fact that policies, often managed through the Non-Real-Time RIC and communicated via the A1 interface, define what actions are permissible for each entity. For example, an application accessing the network might only be authorized to perform diagnostics but not configuration changes. **Granular Authorization** refers to the fact that each O-RAN interface (e.g., O1, A1, E2) implements fine-grained authorization rules to limit what actions can be performed by various entities (e.g., Service Management and Orchestration (SMO) accessing a DU).
4) **Access Control in the O-RAN Architecture:** Access control mechanisms ensure that authenticated and authorized entities can access only the necessary resources [19], [91], [95]. O-RAN includes the following: **Open Interfaces with Access Restrictions** refers to the fact that O-RAN open interfaces (e.g., A1, E2, O1) are secured through role-based and attribute-based access controls. For instance, a DU can only access configuration data from a CU over a secured interface. External systems accessing the SMO are tightly restricted to their assigned domains of control. **Dynamic Access Controls** indicate that access permissions can be dynamically updated based on real-time policies from the Non-RT RIC. For example, an entity's access might be revoked if unusual behavior is detected. Notably, **Zero-Day Threat Mitigation** is supported as access control policies can isolate malicious components in the event of a breach. For instance, if a compromised RU is detected, access controls can prevent it from interacting with the DU.

In this section, we have provided an elaborate discussion on the most recent research that offers diverse strategies for addressing anomaly detection within the O-RAN architecture. O-RAN's reliance on disaggregated, open components necessitates a robust security architecture that incorporates zero trust, authentication, authorization, and access control. By continuously verifying identities, implementing strict access control policies, and leveraging granular authorization, O-RAN can provide a secure and resilient RAN infrastructure while enabling innovation and interoperability. Table XV outlines the latest research endeavors and identifies the specific entity within the O-RAN where each proposed solution is implemented to tackle anomaly detection.

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B. Digital Twinning

The primary aim of the O-RAN architecture is to break down the conventional monolithic base station into modular functional units, enabling flexible deployment and operation. Nonetheless, extending O-RAN principles to incorporate the concept of digital twins, particularly within the telecommunications sphere, necessitates multiple layers of abstraction and integration. Presented below is a conceptual framework outlining an O-RAN architecture tailored for digital twins:

1) **Physical Infrastructure Layer:** This layer encompasses the physical infrastructure of the network, consisting of base stations, antennas, cables, and other hardware components. In the context of DT, each physical asset is replicated virtually within the digital domain.
2) **Data Ingestion Layer:** In this layer, data is gathered from diverse sources within the network infrastructure. This may encompass real-time operational data from sensors embedded in the physical infrastructure, as well as information from network management systems, user devices, and environmental sensors. Continuous updating of the digital twin with this data is crucial to maintaining an accurate reflection of the network's current state.
3) **Digital Twin Representation Layer:** This layer is where the digital representation of the physical network assets is created and maintained. Each asset in the physical infrastructure has a corresponding virtual representation in the digital twin, including its attributes, parameters, and relationships with other assets.
4) **Analytics and Simulation Layer:** Utilizing the data obtained from both the physical infrastructure and the digital twin, this layer executes various analytics tasks and simulations. These tasks may include employing predictive maintenance algorithms to anticipate equipment failures, optimization algorithms to enhance network performance, or simulations to evaluate the effects of configuration changes on network behavior.
5) **Control and Orchestration Layer:** This layer is tasked with orchestrating the network's operation, drawing upon insights derived from the digital twin and analytics layer. It encompasses functions such as network automation, resource allocation, and policy enforcement. The digital twin serves as a tool to simulate various network scenarios and validate changes before they are implemented on the physical infrastructure.
6) **User Interface Layer:** This layer offers interfaces for network operators, engineers, and other stakeholders to engage with the digital twin and retrieve pertinent insights and information. These interfaces may comprise graphical dashboards, visualization tools, and APIs enabling programmatic access to the digital twin data

Through the deployment of an O-RAN architecture tailored for digital twins, telecommunications operators stand to achieve enhanced visibility and management of their network infrastructure, leading to optimized resource allocation and heightened operational efficiency. Furthermore, the modular and open characteristics of O-RAN foster seamless interoperability and integration with diverse network management systems and applications.

Network DT for Traffic Steering: Traffic Steering, a vital use case, entails dynamically routing traffic within the network to enhance performance and regulate network loads. Within the O-RAN framework, the proposed NDT architecture (as shown in Fig. 21) can be harnessed to gather real-time network data and generate a digital model capable of simulating diverse traffic scenarios. Leveraging the NDT enables network operators to make informed decisions regarding traffic flow optimization and network load management, ensuring the desired (QoS/QoE. Moreover, the NDT facilitates the testing of different network configurations and the early identification of potential issues before they manifest in the live network [53].

6G mobile networks anticipate services with diverse requirements, which may fluctuate spatially and temporally. Designing the radio topology based on peak service demands is deemed unnecessary and inefficient. Instead, the network should dynamically adapt to these

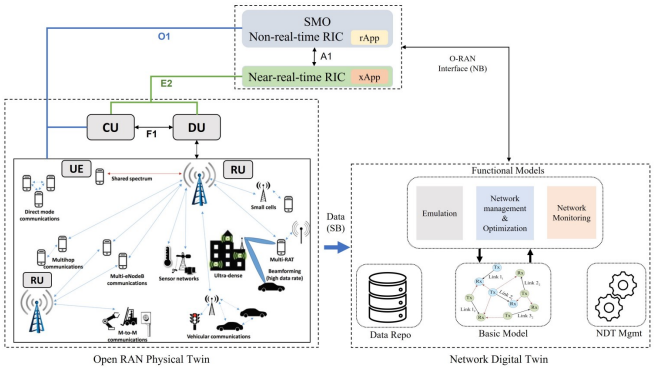


Figure 21. UML sequence diagram for jamming detection [53].

fluctuating demands as needed. Through intelligent RAN moderation, optimal, energy-efficient, and adaptable utilization of radio resources and infrastructure is ensured [54].

Deploying small cells can help alleviate the load on macro cells and cater to a high concentration of users in compact areas like shopping malls, offices, and stadiums. However, maintaining these small cells active at all times, regardless of user density, is not energy-efficient [55]. Implementing a DT of the service area covered by small cells (as shown in Fig. 22), taking into account factors such as mobile devices, their mobility patterns, active applications, and service demands, can be advantageous. This approach aids in forecasting future user density and data demands for small cells, facilitating proactive activation and deactivation of corresponding small cells. Consequently, this minimizes power and spectral resource wastage.

In contrast to static small cell deployments, nomadic nodes (NNs) propose integrating low-power access nodes with wireless backhaul into vehicles such as private cars and taxis. These mobile access nodes assist in extending coverage or enhancing network capacity based on data demand [56]. Traditionally, these nodes are intended to remain stationary during operation, such as when a vehicle is parked, with their availability being random. However, they can be activated or deactivated as needed based on coverage, capacity, or energy requirements. A digital twin of such a system within a designated service area aids in optimal node selection. It considers back-haul link qualities across various available NNs and base stations, models factors like shadowing, multi-path fading, and co-channel interference, and predicts capacity demands and user mobility, among other factors. Moreover, it enables proactive moderation of the movement of these access nodes based on real-time projections of service demands from the DT. Thus, network planning can be conducted dynamically.

Advanced traffic steering methods also contribute to improving the EE of networks by optimizing the active operational duration of access points in the network. A user-centric cell-free massive MIMO network architecture represents a promising aspect of future mobile networks, including 6G or beyond 5G (B5G) [57]. In densely deployed DUs, these systems serve fewer users, with a dedicated serving cluster established explicitly for each user. Such a user-centric no-cell (UCNC) architecture eliminates cell edges, ensuring uniform performance and coverage for users across the network area. However, for users with high mobility, there is often a discrepancy between the estimated channel quality at the time of estimation and the time of data transmission application. This discrepancy is referred to as channel aging [58]. A digital twin of such a distributed antenna system proves useful for dynamically clustering DUs and allocating resources, as depicted in Fig. 22. This DT must account for mobility attributes and user history, and it should incorporate channel aging effects through time-varying modeling of the channel. This modeling relates the temporal auto-correlation function of the channel to user velocity, frequency, propagation geometry, antenna characteristics, and other relevant factors [57]. By doing so, it becomes feasible to explore various methods for RAN moderation and traffic steering.

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Additionally, these methods can be tested and compared in real-time using a DT of the considered solution environment before their implementation in the live network.

Network DT for Radio Resource Management: A radio-aware digital twin (depicted in Fig. 22) can be constructed to encompass the complex intricacies of a radio network. This involves capturing the operations at the Physical and Medium Access Control layers of mobile devices and access points. Additionally, it considers various attributes of these transmitting/receiving nodes, such as their physical coordinates, trajectories, speeds, capabilities, as well as Radio Resource Management procedures, beam patterns, and radio link quality across the entire network. For example, metrics like RSRP or Signal-to-Interference-plus-Noise Ratio (SINR) can be measured on a grid basis and stored as a radio environment map (REM) in a database, which is periodically updated. The primary advantage of utilizing such a DT is its capability for predictive and proactive Radio Resource Managment [59]. The DT empowers the network to anticipate potential errors or failures and devise preemptive solutions accordingly. For instance, if poor link quality is predicted for a mobile device utilizing URLLC services at a certain point in the future (either in terms of time or distance), the network can proactively take measures such as allocating additional resources, switching to different frequency bands, or preparing new beam configurations to mitigate the anticipated issue. It offers advantages over traditional reactive measures within the network, such as re-transmissions, which often struggle to meet QoS constraints, such as delay requirements.

Additionally, a radio-aware digital twin is equipped with an accurate REM that is continuously updated. Propagation prediction techniques like ray-tracing and extended measurements facilitate the construction and maintenance of the REM. In controlled environments like private factory networks, the radio-aware DT possesses knowledge of both current and future positions of mobile devices and traffic patterns, such as sensors transmitting packets at regular intervals. This enables the anticipation of interference patterns and facilitates the estimation of SINR in specific locations, thereby supporting the design of an efficient RRM mechanism [59].

Another benefit of employing a radio-aware digital twin is its ability to forecast link quality between mobile devices and access nodes without the necessity for extensive measurements. By digitally replicating the PHY layer of the network, data-driven models or advanced ray-tracing algorithms can be utilized to accurately estimate channel conditions. This approach reduces the overhead associated with channel estimation, requiring fewer or no channels for direct measurements, consequently enhancing spectral efficiency [59]. Moreover, it can contribute to improved EE, as the power consumed by data-driven models or ray-tracing algorithms is typically lower than that required for comprehensive measurements.

With accurate foresight into future link conditions, it becomes feasible to conduct radio resource management directly on mobile devices for an entire duration, which typically comprises a fixed number of TTIs. This information can be transmitted to the mobile device in a single transmission. Such persistent short-term RRM decisions enable the anticipation of several radio parameters in advance and can include elements like transmit power control, modulation and coding scheme, selection of active Bandwidth-Part (BWP), and allocation of time-frequency resources within a BWP [59]. However, implementing this approach requires the digital twin to possess knowledge of device trajectories and estimates of transmit buffers for devices/access nodes across the specified number of TTIs, ensuring the efficiency of these short-term RRM decisions. This technique is particularly suitable for quasi-deterministic controlled environments like factory automation. By employing these short-term RRM decisions, the overhead associated with conventional RRM procedures can be significantly reduced, as it eliminates the need for periodic measurements performed reactively by mobile devices/access nodes. Given the preceding discussion, it becomes evident how the concept of a DT seamlessly aligns with the principles of openness and intelligence inherent in O-RAN. This section outlines and examines several pertinent use cases that can be implemented in the next-generation 6G network using a DT-based O-RAN architecture. Additionally, a synopsis of the benefits of utilizing DT in O-RAN is presented in Table XVI.

Table XVI
COMPARISON BETWEEN TRADITIONAL AND DT-BASED O-RAN BASED ON DIFFERENT ASPECTS.

Aspect	**Traditional O-RAN**	**DT-based O-RAN**
Network Optimization	Reactive adjustments based on real-time data	Proactive optimization through predictive modeling.
Resource Management	Manual or rule-based resource allocation	Automated and dynamic resource allocation based on predictons.
Performance Monitoring	Limited visibility and periodic monitoring	Continuous real-time monitoring and predictive analytics.
Energy Efficiency	Standard power management strategies	Optimized energy usage through predictive modeling.
Fault Prediction	Limited fault detection capabilities	Early detection of faults and proactive measures.
Service Quality	Limited ability to anticipate and address issues	Improved QoS through proactive measures and optimization.
Scalability	Limited scaliability due to manual intervention	Enhanced scalability through automated and adaptive processes.

C. Vehicle-to-Everything (V2X)

V2X has a dual meaning. Primarily, V2X can be considered as a technology that enables communications between a vehicle and any entity that may affect or be affected by the vehicle. However, V2X can also be seen as encompassing various use cases that leverage this technology to enhance transportation systems.

V2X as a technology refers to the entire suite of communication technologies that allow vehicles to communicate with other vehicles (V2V), infrastructure (V2I), pedestrians (V2P), and networks (V2N). The underlying technology typically involves a combination of cellular communication (C-V2X), Wi-Fi-based communication (DSRC in the U.S.), and potentially other radio communication methods.

V2X use cases include: Safety Applications, such as collision avoidance, emergency vehicle warnings, and pedestrian alerts; Traffic Management, including real-time traffic signal timing adjustments, congestion management, and route optimization; Autonomous Driving, where the V2X technology is used to enhance the situational awareness of autonomous vehicles, allowing them to make safer and more informed decisions and Infotainment, which provide real-time updates on weather, traffic, and nearby amenities to drivers and passengers. Currently, two main strategies for V2X communications exist: Dedicated Short-Range Communications (DSRC) and Cellular Vehicle-to-Everything (C-V2X). Dedicated Short-Range Communications is supported by the IEEE 802.11p standard, while C-V2X is advocated by the LTE or New Radio standards. According to recent recommendations from the 3GPP, vehicular communications at high frequencies, such as millimeter-waves or sub-THz, are pivotal technologies for enabling V2X systems in 6G [60], [61]. However, these frequencies pose reliability challenges, especially in dynamic environments prone to line-of-sight blockages [62].

Due to the highly dynamic nature of environments in V2X communications, link conditions can change rapidly, significantly impacting communication performance. Furthermore, maintaining large-scale connectivity in such dynamic settings poses one of the most challenging aspects in the future 6G V2X systems. Several factors contribute to these challenges, including the high dimensionality of data, high data rates, stringent latency requirements, difficulties with data harvesting and ensuring user privacy, dynamic operator parameters, proprietary vendor data, and irrational bandwidth requirements for training [63]. Addressing these complexities necessitates

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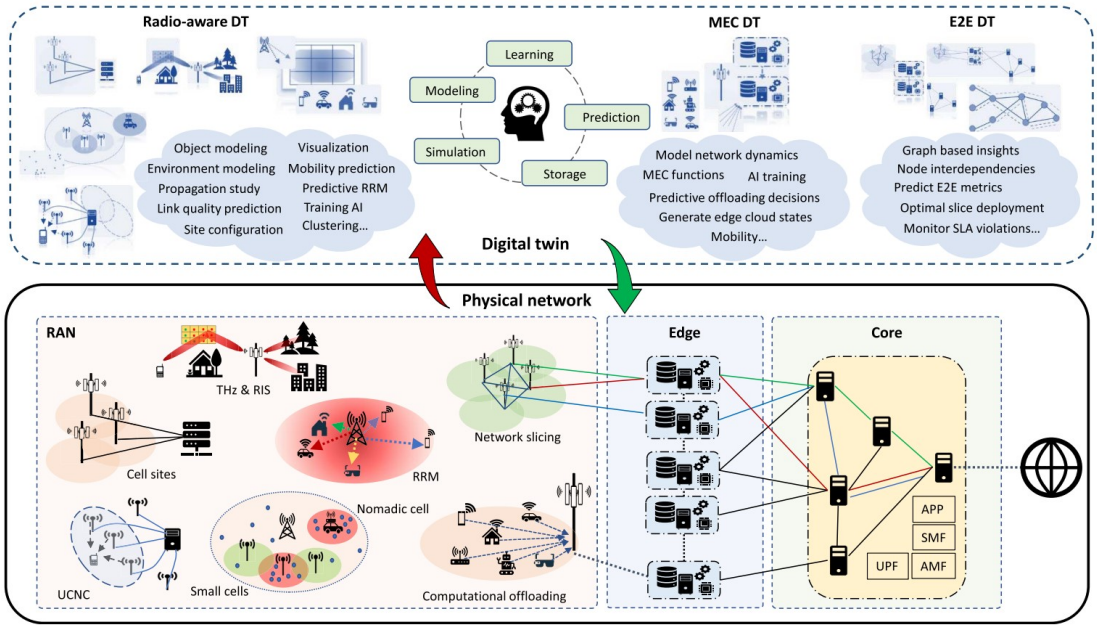


Figure 22. The application of DT technology in various domains and solution areas of the envisioned 6G communication system thereby achieving enhanced system performance and simplifying system operation. [53].

sophisticated orchestration, control, and optimization solutions to ensure seamless service provision in V2X. However, the required data collection and control mechanisms surpass what is currently achievable in traditional RAN deployments.

O-RAN represents the ideal candidate to unlock the potential of V2X through large-scale data collection and dynamic control [64]. In [65], the author introduces the initial user-specific O-RAN TS intelligent handover framework. This framework utilizes the Random Ensemble Mixture approach alongside a Conservative Q-learning algorithm. Additionally, it incorporates a state-of-the-art CNN architecture. These components work together to efficiently allocate a serving base station to each user within the network. The TS xApp, trained using over 40 million data points gathered by ns-O-RAN, operates on the Near-RT RIC platform and manages the ns-O-RAN base stations.

In [66], an O-RAN compliant architecture is proposed for deploying and validating market analysis for the use case of micro-services in V2X communications within a 5G cellular network. The study includes extensive market analysis conducted under dynamic circumstances to identify optimal trends for validation purposes. Additionally, the paper introduces an O-RAN based architecture for making live decisions in near real-time. The optimization process is carried out using free5Gc for Core simulation and the O-RAN Alliance for O-RAN. In [67], a novel integration architecture is advocated to facilitate V2X control within O-RAN. The proposed architecture establishes a low-frequency O-RAN-based control plane to ensure reliable and efficient multi-hop connectivity among Connected Autonomous Vehicles operating at millimeter wave frequencies. To evaluate the technological feasibility of this integrated architecture, the authors extend the capabilities of the current NS-3 modules, creating a simulation framework for experimenting with O-RAN-enabled V2X systems. This integrated framework is utilized to demonstrate the benefits in addressing a specific V2V challenge: link and relay selection. The authors of [64] centered their focus on leveraging the O-RAN architecture for V2X communications. They deliberated on how the O-RAN architecture holds promise in delivering a more adaptable, scalable, and cost-efficient solution when contrasted with existing V2X systems. Furthermore, they examined integration points and challenges within open architectures, which could pave the way for exploring new research directions.

A new age of mobility, where cars may smoothly connect with infrastructure, pedestrians, and each other, is heralded by O-RAN-based V2X. In addition to improving safety, this synergy optimizes traffic flow and lowers emissions and congestion. Through the utilization of O-RAN's adaptability and scalability, V2X applications may adjust to a variety of settings, including busy cities and isolated routes. In order to fully realize the potential of O-RAN-powered V2X, cooperation, standardization, and regulatory alignment must come first. We can realize the full potential of V2X technology and build safer, smarter, and more sustainable transportation systems for future generations by establishing an ecosystem where players from all industries can develop and work together.

Table. XVII outlines various use cases where O-RAN plays a crucial role in V2X communications, ranging from V2V and V2I communication to interactions with pedestrians, networks, and the power grid.

D. Massive MIMO

Beamforming has been established with tremendous advantages to address several communication issues, including improving radio connection precision, throughput, number of parallel connections in a given cell area, and transmission energy savings. Beamforming is very useful in mmWave transmission for enhancing signal-to-noise ratios (SNR) by directly targeting user groups, primarily indoor coverage; also, beamforming can eliminate interference amongst UE singles in high-density situations. Due to all these advantages, there are now additional obstacles for developers, testers, and antenna designers to overcome, such implementation costs, design complexity, and the need for testing in a real physical setting. Nonetheless, the 3GPP and O-RAN standards have taken into account all of these difficulties. Split choices, which 3GPP offered, made it easier for the RAN's designers to accomplish fixed implantation. In front-haul, where the C-plane communication transmits the beamforming data in multiple beamforming techniques, O-RAN adopted the Split 7-2x (as shown in Fig. 9) option, which introduces simplified communication interference between O-RAN units. All beamforming technologies—digital, analog, or hybrid—have been supported by O-RAN in the front-haul interface.

In the front-haul interface, O-RAN implemented Spit Option 7-2x, which necessitated a large bandwidth. The authors in [68] propose use Category B to carry out the precoding in O-RU in order to get around this problem. Because each O-RU will compute its own beamforming matrices, this approach will result in inter-RU interferences in O-RU.

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Table XVII
O-RAN IN V2X

Use Case	Description
V2V Communication	O-RAN facilitates direct communication between vehicles, enabling real-time exchange of data such as position, speed, and trajectory for collision avoidance.
V2I Communication	O-RAN enables vehicles to communicate with infrastructure such as traffic lights and road signs, optimizing traffic flow and enhancing safety.
V2P Communication	O-RAN extends communication to pedestrians, alerting vehicles of their presence and intentions, enhancing safety in urban environments.
V2N Communication	O-RAN enables vehicles to connect with networks, accessing cloud-based services for navigation, traffic updates, and remote diagnostics.
V2G Communication	O-RAN facilitates vehicle-to-grid communication, allowing electric vehicles to interact with the power grid for optimized charging and energy management.
V2X Security	O-RAN provides a secure framework for V2X communication, implementing authentication, encryption, and integrity checks to prevent cyber threats.

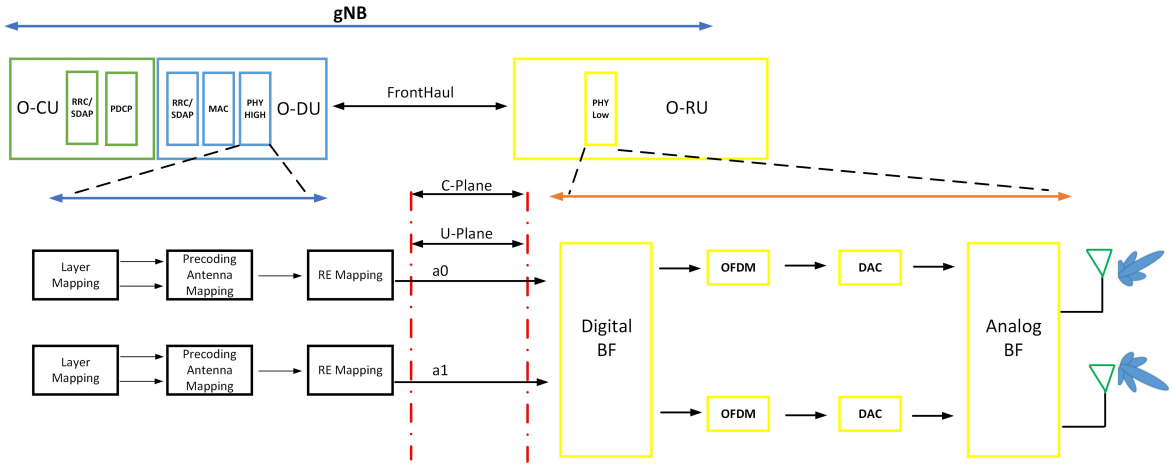


Figure 23. Hybrid beamforming block diagram in O-RAN.

Therefore, it has been proposed to create a precoding beam in O-RU and construct the beamforming matrices in O-DU using the predicted channel information. With the precoding now integrated into the O-RU unit, they suggest using the zero-forcing (ZF) algorithm as a potential means of minimizing interference in received signals while preserving beamforming in a single unit. As a result, hardware complexity and implantation costs will go down [68]–[71].

Split Option 7-2x supports both precoding in O-DU (Category A) and O-RU (Category B), and it implements functions up to the resource element mapping in the O-DU. Category A: O-DU handles the precoding (digital beamforming) function, making the O-RU design straightforward (as shown in Fig. 23). Spatial streams will be sent in the downlink via the front-haul interface. When it comes to low-frequency and latency-insensitive services, Category A works well. Category B: O-RU performs precoding (analog and digital beamforming), which complicates the O-RU architecture. Spatial layers will be carried by the front-haul interface. This category's benefit is its lower downlink throughput, which is made possible by modulation compression in the downlink, which sends just the bits that correspond to the constellation points. High-frequency and high-traffic networks are well-suited for Category B networks [69].

1) O-RAN Beamforming Methods: O-RAN has suggested the following techniques to attain satisfactory beamforming performance:

1) **Indexing beamforming (predefined-beam beamforming)**- Based on BeamID, O-DU notifies O-RU about the type of beamforming, and O-RU records this information in a database of beam indexes. This table includes beam weights; the BeamID is used as a pointer to a pre-defined beamforming weight vector. The beam weights can be either frequency domain (digital beamforming), time-domain (analog beamforming), or both in hybrid beamforming. In order to identify which beam should be applied, BeamID is only a weight vector value that is transferred between O-DU and O-RU.
2) **Dynamic beamforming based on weights in real time.** With this approach, the beamforming weights produced in real time by the O-DU are sent to the O-RU to be linked to the data of a particular user. If the beamforming weights remain constant for a while, a beam index can be set and used in later messages. Weight-based dynamic (digital and analog) beamforming and weight-based dynamic hybrid beamforming are the two implementation scenarios for this technology; in the first, the beamforming weights may be changed in real time using the BeamID value linked to the weights. Two sub-cases are examined in the second case: (1) the frequency-domain weights and the time-domain weights may be updated in real-time; and (2) the frequency-domain weights and the time-domain beams may be updated in real-time but remain constant. It should be noted that time domain weights are applied to the time domain IQ data per OFDM symbol, while frequency domain weights are applied to the frequency domain IQ data per section.
3) **Beam Attributes Method (Attribute-Based Dynamic Beamforming):** The primary limitation of the preceding method is its reliance on additional processing capability within the O-RU to adjust transmission based on the beam weights provided by the O-DU. Therefore, an adaptation of the previous approach has been suggested, leading to the development of the Beam Attributes Method. In this modification, the responsibility for generating beamforming weights shifts to the O-RU, utilizing a predefined set of beam attributes communicated by the O-DU through the control plane. These attributes encompass (1) azimuth and elevation, (2) beam width (narrow or wide), (3) side

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lobe suppression, and (4) beam weights calculated by the RU. One consequence of this approach is the exclusion of hybrid beamforming in the Beam Attributes Method. This is because the weights generated by the O-RU are either frequency-domain or time-domain, but not a combination of both.

4) **Channel Information Method (Channel-Information-Based Beamforming):** This approach relies on digital beamforming, known for its efficacy in high-traffic communication environments due to its capacity and flexibility, offering steerable beams. In digital beamforming, each antenna possesses a dedicated RF signal and path, enabling a high number of dynamically transformable beams without requiring hardware modifications. In this method, the O-RU generates beam weights utilizing channel information provided by the O-DU. The algorithms employed can synthesize a drive signal to optimize various beam characteristics. However, implementing precoding in the RU can result in inter-RU interferences within the O-RU. To address this issue, algorithms such as the zero-forcing algorithm (ZF) can be utilized to mitigate interference in received signals.

The objective of paper [72] is to introduce a Beam Management algorithm that leverages location-specific information stored in a REM. By utilizing received power maps and user mobility patterns, the algorithm aims to optimize the Beam Management process in terms of Radio Link quality. This optimization is achieved through the application of the Policy Iteration method under various objective functions, such as maximizing received power or minimizing beam reselections, while simultaneously avoiding radio link failures. The proposed solution adheres to the principles of the Open RAN architecture, ensuring its compatibility and feasibility for practical implementation in real-world scenarios. The authors of paper [73] present an integration of the Internet of Energy into 5G infrastructure. They discuss how the requirements of the Internet of Energy can potentially be met by leveraging the capabilities of the 5G network. The authors address the challenge of establishing a communication network capable of implementing tailored solutions for the Internet of Energy. They conclude that adopting an O-RAN implementation offers a viable approach, as it reduces costs by streamlining certain aspects of the network and enhances the potential for various applications by increasing deployment flexibility within the network. However, the O-RAN approach also has its limitations, which must be addressed to enable its use in Internet of Energy applications. One notable limitation is the feasibility of deploying beamforming mechanisms based on MIMO antennas within this framework. To address this challenge, they introduced the concept of zero-forcing within O-RAN, aiming to alleviate traffic congestion in the fronthaul. Our findings indicate that zero-forcing and Minimum Mean Square Error (MMSE) techniques effectively mitigate interferences, offering a promising solution for deploying beamforming in O-RAN implementations.

In paper [74], the authors devised a learning-driven Pilot Assignment strategy tailored for the decentralized cell-free mMIMO system embedded within the O-RAN architecture (as shown in Fig. 24). They opted for O-RAN as a pragmatic system framework, benefiting from its delineated network functions and multi-timescale control loops, which effectively govern the operation of their strategy. Upon formulating the Pilot Assignment quandary and crafting a corresponding Markov game model, they engineered a PA algorithm grounded on the Multi-Agent Deep Reinforcement Learning (MA-DRL) framework. Additionally, they devised a Codebook Search mechanism that expedites the learning-based PA process, minimizing Mean Squared Error (MSE) without introducing substantial additional complexities. In comparison to prevailing benchmarks, their approach exhibited satisfactory performance metrics concerning both channel estimation MSE and computational scalability. Moreover, in contrast to many prevailing PA tactics, their strategy does not necessitate any prior channel information.

In [75], the authors introduced $O-M^3$, a real-time scheduler designed for a multi-cell MIMO system operating within the O-RAN architecture. $O-M^3$ is engineered to facilitate cooperative

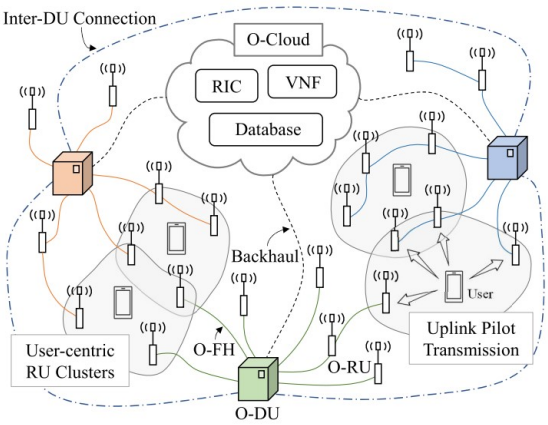


Figure 24. A decentralized CFmMIMO system realized in O-RAN [74].

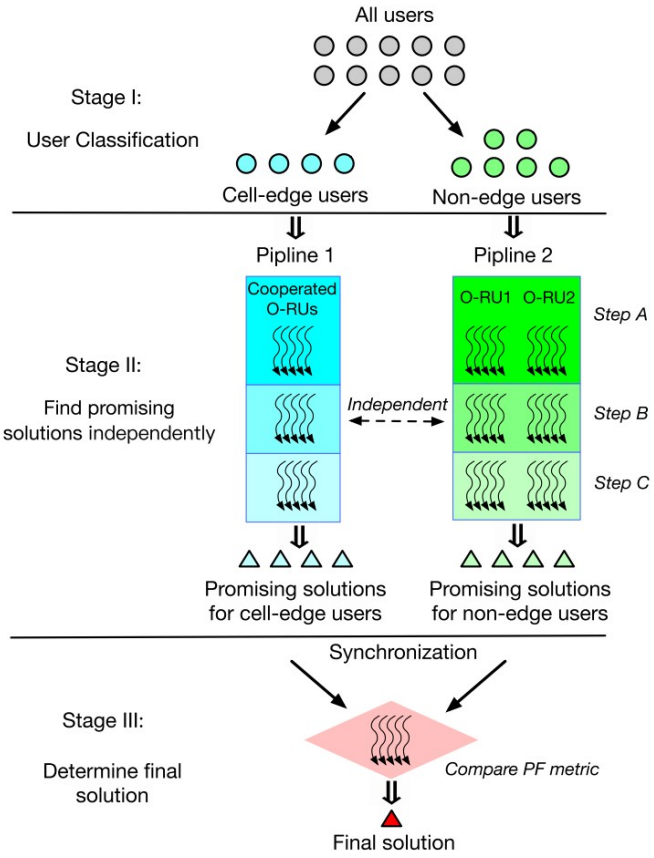


Figure 25. A flow chart for $O-M^3$ [65].

signal processing across multiple cells at an O-DU (as shown in Fig. 25). Specifically, it supports joint transmission and optimizes resource block (RB) allocation, modulation and coding scheme (MCS) assignment, and beamforming matrices for all users across the O-RUs. Crucially, $O-M^3$ delivers a solution within the stringent 1 ms timeframe prescribed by the frame structure defined in 5G NR. To meet the demanding real-time requirements, the authors devised a novel multi-pipeline design that capitalizes on extensive parallelism. Through comprehensive experimentation, they demonstrated that $O-M^3$ consistently achieves scheduling solutions within the 1 ms deadline across all tested scenarios. Furthermore, they illustrated that $O-M^3$ substantially enhances user data rates by implementing joint transmission among multiple cells.

In summary, the integration of Massive MIMO technology into the O-RAN framework offers substantial potential, especially in the realm of beamforming applications. Leveraging the adaptability

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Figure 26. Major O-RAN Challenges.

and openness of O-RAN, operators can implement Massive MIMO systems more effectively, resulting in enhanced network performance and user satisfaction. The case study of beamforming highlights the capabilities of Massive MIMO within O-RAN environments. Techniques like zero-forcing and minimum mean square error (MMSE) tackle issues such as interference and congestion in fronthaul traffic, thereby improving the overall efficacy of beamforming strategies in O-RAN.

Moreover, the emergence of novel scheduling algorithms, as discussed in the literature, emphasizes the significance of real-time optimization in multi-cell MIMO systems. These algorithms enable collaborative signal processing across multiple cells, leading to notable enhancements in user data rates while adhering to the strict timing constraints of 5G NR. In essence, the convergence of Massive MIMO and O-RAN presents an enticing opportunity to elevate network performance, enhance flexibility, and drive down costs in the realm of next-generation wireless networks. As ongoing research and development efforts advance, we anticipate further innovations that will shape the landscape of wireless communication infrastructure.

VIII. Challenges

This section discusses the major challenges of O-RAN deployment, as illustrated in Fig. 26. The challenges identified are classified as follows: 1) Examining Again, 2) Standards Adherence, 3) Sustaining Independence, and 4) Over Innovation.

A. Monitor and Update [89]

This challenge relates to the complexity management and continuous optimization of the O-RAN architectures. The issue of complexity and the need for constant examination of systems as they evolve is well-recognized in O-RAN literature. Here is a breakdown of the challenges with references supporting each point:

1) **Complexity Management:** Managing the increasing complexity of the O-RAN architectures as they evolve and incorporate new technologies. [76] discusses the growing complexity of O-RAN architectures and the need for effective management strategies.
2) **Interoperability Issues:** Ensuring seamless interoperability between components from different vendors to maintain the openness of O-RAN deployments. [77] highlights the importance of interoperability standards in O-RAN deployments and the challenges associated with achieving seamless integration.
3) **Optimization:** Continuously optimizing network performance, resource allocation, and traffic management processes to adapt to changing conditions and demands. [78] discusses the challenges and strategies for optimizing O-RAN deployments to meet evolving requirements.

By addressing these challenges through collaboration, innovation, and continuous improvement, stakeholders can overcome barriers to O-RAN adoption and unlock its full potential in transforming the telecommunications industry [89].

B. Standard Adherence [90]

This challenge highlights difficulties in maintaining compliance with evolving and sometimes non-unified standards, a well-documented issue in the development of open systems like O-RAN. The need for interoperability and adherence to standards is often referenced, as standards continue to evolve in the rapidly changing landscape of telecom technologies:

1) **Compliance Verification:** Ensuring compliance with standards and specifications throughout the lifecycle of O-RAN deployments, including procurement, deployment, and operation. [79] explores the challenges of verifying compliance with standards in O-RAN deployments and strategies to ensure adherence.
2) **Evolving Standards:** Adapting to evolving standards and specifications as the O-RAN ecosystem matures and new technologies emerge. [80] discusses the challenges of keeping pace with evolving standards in O-RAN deployments and the importance of flexibility and agility.
3) **Vendor Lock-In Mitigation:** Mitigating the risk of vendor lock-in by promoting open standards and interoperability to foster a diverse and competitive vendor ecosystem. [81] examines the challenges associated with vendor lock-in in O-RAN deployments and strategies to promote open standards and interoperability.

Addressing these challenges requires collaboration among industry stakeholders, standardization bodies, and regulatory authorities to develop and adhere to open and interoperable standards that drive innovation and ecosystem growth in the O-RAN space [90].

C. Interoperability Support [91]

Refers to avoiding vendor lock-in and promoting the independence of operators from proprietary systems, which is a core motivation behind the O-RAN initiative. Maintaining interoperability while fostering a competitive ecosystem is a widely recognized challenge:

1) **Reducing Vendor Dependence:** Avoiding dependence on a single vendor or supplier for O-RAN components, ensuring operators have the flexibility to choose best-of-breed solutions and prevent vendor lock-in. [82] discusses the challenge of reducing dependency on incumbent vendors and strategies for fostering a diverse supplier ecosystem.
2) **Maintaining Interoperability:** Ensuring interoperability between components from different vendors while maintaining independence from proprietary solutions and closed ecosystems. [83] examines the interoperability challenges faced by operators deploying O-RAN solutions and the importance of maintaining independence from vendor-specific technologies.

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3) **Integration Flexibility:** Balancing the need for integration flexibility with the risk of becoming overly dependent on integrators or system integrators for O-RAN deployments. [84] explores the challenges of achieving integration flexibility while maintaining independence from integrators and system vendors.

Addressing these challenges requires collaboration and coordination among industry stakeholders, including network operators, vendors, standardization bodies, and regulatory authorities. By collectively addressing technical, operational, and organizational challenges, the O-RAN ecosystem can overcome barriers to adoption and realize its potential to drive innovation, competition, and efficiency in the telecommunications industry [91].

D. Innovation Risks [92]

Deals with the risks of over-complicated O-RAN deployments through excessive innovation. Issues like cost overruns, operational complexity, and technology fragmentation are well-cited in the literature, particularly as O-RAN evolves to accommodate next-generation technologies:

1) **Cost Overruns:** Over-innovating O-RAN deployments without careful consideration of cost implications can lead to budget overruns, jeopardizing the financial viability of projects and hindering widespread adoption. [85] discusses the importance of balancing innovation with cost considerations in O-RAN deployments to avoid budget overruns and ensure financial sustainability.
2) **Operational Complexity:** Implementing overly innovative features or technologies in O-RAN deployments may increase operational complexity, requiring specialized skills, resources, and support mechanisms that operators may struggle to maintain. [86] examines the operational complexities associated with over-innovating O-RAN deployments and the implications for network management and support.
3) **Technology Fragmentation:** Over-innovating O-RAN deployments may result in technology fragmentation, with operators adopting disparate solutions and standards that hinder interoperability and scalability. [87] discusses the risks of technology fragmentation resulting from over-innovation in O-RAN deployments and strategies to promote interoperability and standardization.

To mitigate the challenges of over-innovation, operators should adopt a balanced approach that prioritizes innovation aligned with strategic objectives, interoperability, cost-effectiveness, and operational simplicity in O-RAN deployments [92].

E. Potential Challenges in Implementing O-RAN in 6G:

1) Interoperability and Standardization:
   Challenge: While O-RAN aims to foster interoperability between different vendors' equipment, achieving seamless integration across a highly diverse ecosystem remains a significant challenge. The absence of universally accepted standards could lead to compatibility issues, slowing down deployment and potentially undermining performance.
   Solution: To address this, the industry must accelerate the development and adoption of standardized interfaces and protocols. Collaboration among global standardization bodies, vendors, and operators is crucial to ensure that O-RAN components can work together harmoniously. Additionally, extensive testing and validation frameworks should be established to verify interoperability across different vendors' products before deployment in 6G networks.
2) Security Vulnerabilities:
   Challenge: The open and disaggregated nature of O-RAN, while advantageous for flexibility, also increases the attack surface, making the network more vulnerable to cyber threats. Ensuring robust security in such a complex and dynamic environment is a major concern.

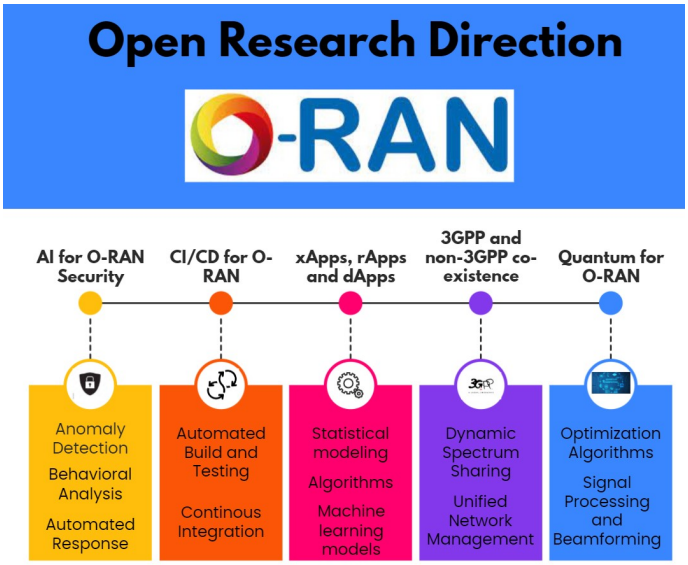


Figure 27. Open Research Directions for Open RAN.

   Solution: Addressing security challenges requires a multi-layered approach, integrating security at every level of the O-RAN architecture. AI-driven threat detection and response mechanisms can be employed to identify and mitigate risks in real-time. Moreover, establishing strict security protocols and best practices, along with regular security audits, will be essential to protect the network from emerging threats as we transition to 6G.
3) Performance Optimization:
   Challenge: While O-RAN offers the flexibility to deploy and optimize network functions, ensuring that these functions perform optimally under the high demands of 6G, such as ultra-low latency and massive connectivity, can be challenging. The disaggregated nature of O-RAN might lead to inefficiencies or performance degradation if not managed carefully.
   Solution: Implementing advanced AI and machine learning algorithms within the RAN Intelligent Controller (RIC) can enable dynamic, real-time network optimization. These technologies can predict and adapt to network conditions, ensuring that performance targets are consistently met. Furthermore, close collaboration between vendors and operators during the design and deployment phases can help tailor O-RAN solutions to the specific requirements of 6G, mitigating potential performance issues.
4) Operational Complexity:
   Challenge: The flexibility and modularity of O-RAN introduce a higher level of complexity in terms of deployment, management, and troubleshooting. Operators will need to manage a more intricate network with components from multiple vendors, which could lead to increased operational costs and complexity.
   Solution: To manage this complexity, the development of automated management and orchestration tools is essential. These tools should be capable of handling the lifecycle management of O-RAN components, from deployment to maintenance, with minimal human intervention. Training and upskilling of the workforce will also be necessary to equip operators with the expertise required to manage and optimize O-RAN-based 6G networks effectively.

In this section, we have examined the most recent obstacles related to O-RAN. Additionally, we have provided recent references to enable the research community explore these challenges further and identify potential solutions.

IX. Open Research Directions

This section presents several open research directions for O-RAN, as illustrated in Fig. 27. Following an extensive study, the authors

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of this paper have identified the following five main open research directions: 1) AI for O-RAN Security, 2) CI/CD in O-RAN design, 3) xApps, Apps and dApps, 4) 3GPP and non-3GPP co-existence and 5) Quantum for O-RAN. They are discussed next in more details.

A. AI for O-RAN Security

Security is a fundamental concern for O-RAN, given its open and disaggregated architecture. Section VII of this survey discusses the challenges introduced by the multi-vendor ecosystem and the open interfaces, which increase the attack surface. The complexity of securing a distributed, multi-vendor environment demands more advanced, AI-driven security solutions [9].

Current solutions rely heavily on traditional security mechanisms, but these may not be adequate to detect sophisticated and dynamic threats. AI/ML techniques can offer real-time threat detection, anomaly detection, and adaptive security mechanisms, as discussed earlier in the context of security challenges. However, further research is needed to develop AI models that can scale to the distributed nature of O-RAN, remain computationally efficient, and handle high traffic volumes in real-time. Integrating AI-driven security as part of the RIC can provide enhanced, automated protection across multiple layers of the network.

Motivation: Increased attack surface due to open interfaces and multi-vendor environments. Need for real-time, AI-driven security to detect complex cyber threats. Requirement for scalability and efficiency in large-scale 6G deployments.

B. CI/CD for O-RAN

As outlined in Sections V and VI, the dynamic and flexible nature of O-RAN requires continuous updates and optimizations to its software and hardware components. This creates a critical need for CI/CD pipelines that ensure efficient and secure software rollouts across a disaggregated network infrastructure. Currently, the integration of CI/CD in O-RAN environments is still in its infancy, and more research is required to standardize the processes of automated testing, continuous deployment, and software orchestration without disrupting network operations [13].

The integration of CI/CD in 5G O-RAN environments is crucial for enabling agile development, rapid deployment, and continuous improvement of network functionalities. However, CI/CD workflows in such complex and dynamic environments must be closely tied to robust monitoring and observability frameworks to ensure operational excellence. Despite the benefits, the discussion often lacks depth in addressing the pivotal role of these frameworks, particularly when compared to the insights from existing studies such as [109]. This survey highlights how observability in 5G RAN systems, through real-time metrics collection, fault detection, and anomaly identification, is essential for maintaining the reliability and adaptability of CI/CD processes.

In the context of O-RAN, CI/CD practices must be supported by advanced observability mechanisms tailored to disaggregated and heterogeneous architectures. As outlined in the literature, observability frameworks like O-RAN's FCAPS (Fault, Configuration, Accounting, Performance, and Security) interfaces and monitoring through the Open Fronthaul M-plane are instrumental in providing real-time insights into network behavior. These tools can detect issues early, enabling rapid rollback or updates through CI/CD pipelines while minimizing service disruptions. Furthermore, the integration of AI and ML for predictive analytics and intelligent automation of these pipelines ensures dynamic resource scaling, real-time optimization, and improved fault recovery, all of which align with the stringent performance and reliability requirements of 5G and beyond [101].

To strengthen CI/CD implementations, the inclusion of monitoring standards like KPIs for energy efficiency, latency, and fault rates across RUs, DUs, and CUs is essential. Such metrics, combined with comprehensive observability, support not only CI/CD processes but also the broader goals of O-RAN, including energy efficiency and low-latency communication. Thus, enhancing CI/CD discussions with

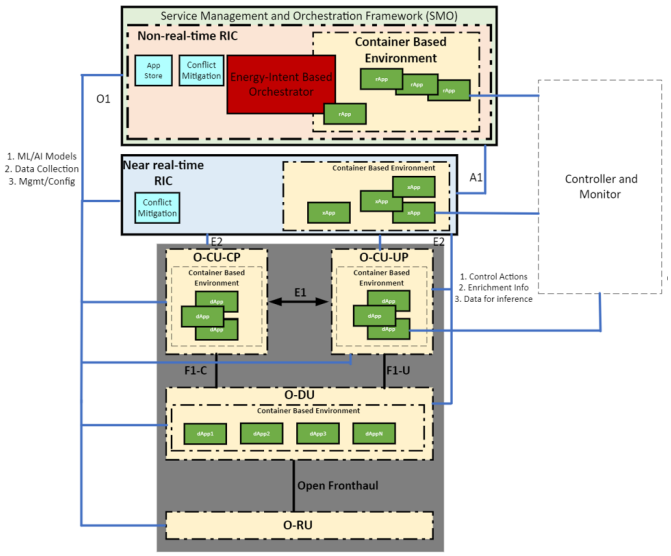


Figure 28. xApps, rApps and dApps Placement in O-RAN Architecture [12].

references to existing frameworks and challenges in monitoring and observability provides a more substantiated foundation for leveraging these practices effectively in 5G and future networks.

The challenges here relate to managing multi-vendor software stacks and ensuring that updates do not introduce vulnerabilities or impact network performance. Research should also focus on how to achieve zero-downtime updates and real-time performance monitoring during software deployments, which are vital for ensuring reliability and availability in O-RAN.

Motivation: O-RAN's modular and disaggregated nature demands frequent updates, making CI/CD critical for network reliability. The complexity of managing software stacks across multiple vendors requires standardization of CI/CD processes. Zero-downtime deployment and real-time monitoring are essential for maintaining network availability.

C. xApps, rApps and dApps for O-RAN

Section VI discussed the role of the RAN Intelligent Controller (RIC) in managing and optimizing O-RAN components (as shown in Fig. 28). Specifically, the survey introduced the concept of xApps (near-real-time applications) and rApps (non-real-time applications), which are critical for implementing network automation, resource management, and policy enforcement. The development of these applications is a key area of future research, particularly for ensuring they can support emerging 6G use cases such as autonomous networks, dynamic resource allocation, and predictive maintenance [12]. Furthermore, dApps, which leverage blockchain technology, could introduce trust and transparency in multi-vendor O-RAN environments. However, the use of dApps in O-RAN is still largely unexplored and presents opportunities for research in areas like secure network management, trustless transactions, and decentralized network control.

Motivation: xApps and rApps are essential for network automation and optimization, but their scalability and integration with 6G technologies need further research. dApps could address trust and transparency issues in O-RAN, especially in multi-vendor ecosystems. Research is needed to optimize the deployment of these applications in real-time and non-real-time network environments.

D. 3GPP and Non-3GPP Co-existence

A major challenge discussed in Section V is the integration of 3GPP-compliant technologies with non-3GPP solutions, particularly in O-RAN deployments. With the advent of 6G, there will be a need to integrate heterogeneous networks, including those based on Wi-Fi, satellite communication, and IoT protocols, alongside traditional

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cellular networks. O-RAN's open interfaces offer an opportunity for seamless co-existence, but more research is needed to explore how these technologies can interoperate without introducing latency or compromising security [14].

Research in this area should focus on interoperability frameworks, spectrum sharing mechanisms, and cross-network authentication protocols. The coexistence of 3GPP and non-3GPP technologies will be critical for ensuring ubiquitous coverage and seamless mobility across different communication networks.

Motivation: Ensuring interoperability between 3GPP and non-3GPP technologies is essential for heterogeneous network integration. Need for seamless mobility and cross-network authentication in O-RAN, particularly as 6G integrates more diverse technologies. Research is required to address challenges such as latency, security, and resource allocation in multi-standard environments.

E. Quantum for O-RAN

Quantum computing (QC) is mentioned as a research challenge in the context of O-RAN for 6G because of its potential to revolutionize areas like security, optimization, and processing efficiency, which are critical to the architecture and technologies discussed in the paper. Specifically, QC can address computationally intensive tasks, such as optimizing large-scale beamforming algorithms in massive MIMO systems or enhancing predictive capabilities in AI/ML models used for resource allocation and anomaly detection. Furthermore, QC holds significant promise in improving cryptographic techniques, such as quantum key distribution (QKD) [110], [111], which could strengthen the security of O-RAN systems by ensuring unbreakable encryption methods. In the disaggregated and multi-vendor nature of O-RAN, where security and performance are paramount, QC-based solutions can enable secure communication and efficient network management. While still an emerging technology, its integration into O-RAN aligns with the forward-looking scope of 6G, making it an essential topic for future exploration despite its current nascent state. This paper highlights QC as a promising but underexplored avenue, recognizing its relevance to the scalability and security challenges of next-generation networks [112].

The potential of QC to revolutionize telecommunications is significant but remains largely unexplored, as noted in Section VII. QC can address complex problems in resource allocation, security, and optimization that are computationally prohibitive for classical computers. For O-RAN, quantum algorithms could be applied to optimize network routing, solve complex beamforming problems in massive MIMO systems, and enhance cryptographic security with quantum-safe encryption methods [14]. However, integrating QC into O-RAN presents several challenges, including the need for quantum-compatible infrastructure and overcoming current hardware limitations. Research must explore how to make quantum resources accessible in real-time, distributed O-RAN environments and ensure compatibility with classical network functions.

Motivation: QC can significantly improve O-RAN's ability to solve complex optimization and security problems, particularly in massive MIMO and beamforming scenarios. Quantum-safe encryption is critical for future-proofing O-RAN against quantum-enabled attacks. The integration of QC into O-RAN requires significant research to develop compatible infrastructure and algorithms.

In this section, we have presented the latest O-RAN research directions which can help researchers focus on and develop new algorithms for advancement of O-RAN. The O-RAN Alliance has introduced The Next Generation Research Group (nGRG) which emerged as a research task force established by the O-RAN Alliance in June 2022. Comprising members representing major mobile operators, vendors spanning the telecom, IT, and DT sectors, as well as esteemed research institutions and universities globally, the nGRG is charged with a mission to harness industry and academic research endeavors related to 6G technology and beyond. This mission encompasses determining the evolution of the O-RAN to accommodate forthcoming 6G advancements, taking into account regional research endeavors, ITU-R specifications, and developments within 3GPP. The nGRG is entrusted with defining the O-RAN Alliance's research agenda and outlining key priorities concerning the 6G technology. Its primary objectives include ensuring the sustainability of O-RAN from the transition from 4G/5G to 6G and beyond, and harmonizing the trajectory of 6G technology to avert compatibility issues with other standards development organizations.

F. Comparative Analysis of Key Research Directions

To guide future research, we summarize key open directions with their advantages, disadvantages, and the trade-offs involved in Table. XVIII, Table. XIX, and Table. XX.

G. Recommendations and Future Directions

Enhanced AI/ML Integration: Develop hybrid AI models combining supervised, unsupervised, and reinforcement learning techniques to improve network optimization and security.

Quantum Readiness: Invest in research for quantum-safe cryptographic algorithms and explore quantum applications for resource optimization in O-RAN.

Standardization Efforts: Foster collaboration between industry consortia and standardization bodies to address interoperability challenges and set benchmarks for emerging solutions.

Sustainable Networks: Focus on energy-efficient designs by combining dynamic spectrum allocation with sleep-mode strategies for idle components.

Blockchain Scalability: Explore lightweight blockchain solutions optimized for real-time operations in O-RAN environments.

The transition from Section VIII, which addresses the significant challenges of implementing O-RAN in the context of 6G, to Section IX, which outlines the open research directions, highlights the critical need for continuous innovation and targeted solutions to overcome existing limitations. Section VIII emphasizes hurdles such as interoperability, security vulnerabilities, performance optimization, and operational complexity within O-RAN architectures. These challenges underscore the intricate balance between openness, flexibility, and network efficiency, which becomes even more critical as we move toward the advanced requirements of 6G networks. Building upon these identified challenges, Section IX explores potential research opportunities aimed at addressing these concerns, including the integration of AI-driven security mechanisms, efficient CI/CD pipelines, advanced xApps and dApps for network optimization, and the coexistence of 3GPP and non-3GPP technologies. This progression creates a cohesive narrative, bridging current limitations with forward-looking solutions, and establishes a foundation for next-generation O-RAN deployments in the 6G era.

X. Conclusions

While 6G networking is currently in its conceptual stage, research and development endeavors are in their nascent phases. Nevertheless, insights from the evolution of O-RAN can offer valuable lessons that could guide the development of 6G networking architecture, delineate potential use cases, and highlight unresolved issues:

1) **Flexibility and Interoperability:** O-RAN's emphasis on flexibility and interoperability, achieved through the disaggregation of hardware and software components, offers valuable lessons for 6G networking. A similar approach in 6G could foster greater innovation, competition, and customization, leading to more efficient and adaptable network architectures.
2) **Virtualization and Cloud-Native Architectures:** O-RAN's promotion of virtualization and cloud-native architectures has facilitated the deployment of network functions as software-based instances on standard hardware. Extending these principles in 6G could support ultra-low latency, massive connectivity, and distributed computing capabilities required for emerging use cases like augmented reality (AR), virtual reality (VR), and tactile internet.

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Table XVIII
COMPARATIVE ANALYSIS OF KEY RESEARCH DIRECTIONS

Research Direction	Approach / Technology	Advantages	Disadvantages	Trade-offs
AI/ML for Security in O-RAN	AI-based threat detection	Real-time monitoring, improved anomaly detection	Dependence on robust data quality and volume, risk of adversarial attacks	Balances security improvements with computational complexity
CI/CD for Dynamic Deployments	Automation pipelines for xApps/rApps	Accelerates innovation, ensures consistency in software updates	High initial setup cost, potential for deployment errors	Speeds development cycles but increases operational challenges
Quantum Computing for O-RAN	Quantum cryptography and processing	Unbreakable encryption, faster processing for resource-intensive applications	Immaturity of technology, high implementation cost	High potential but requires long-term investment
Energy-Efficient Protocols	Dynamic spectrum and resource scaling	Reduced energy consumption, greener networks	Risk of under-provisioning, dependency on precise traffic forecasts	Reduces energy costs but may affect user experience
Interoperability in Heterogeneous Networks	Standardized interfaces	Seamless integration of 3GPP and non-3GPP technologies	Difficult to standardize across diverse vendors	Enhances connectivity but introduces standardization challenges
Blockchain for O-RAN Security	Decentralized ledger-based access control	Enhanced data integrity, transparency	Latency overhead, scalability issues for real-time applications	Improves security but increases transaction costs

Table XIX
TRADE-OFFS IN RESEARCH APPROACHES

(A) AI/ML for O-RAN Security		
Aspect	**Advantages**	**Disadvantages**
Threat Detection	Real-time response to evolving cyber threats	May struggle with novel threats
Scalability	Scalable with data volume	High resource requirements for training and inference
Autonomy	Reduces human intervention	Risk of false positives and negatives
(B) CI/CD Implementation		
Agility	Faster software updates and deployments	Increased complexity in managing pipelines
Consistency	Reduces human error in repetitive tasks	Debugging automated processes can be challenging
Scalability	Supports large-scale networks	Initial cost of automation infrastructure is high
(C) Quantum Computing in O-RAN		
Processing Power	Can handle complex optimization tasks	Technology is still in nascent stages
Security	Enhances encryption and secure communications	Hardware remains expensive and resource-intensive
Scalability	Theoretically scalable	Practical implementation is limited

3) **AI and Machine Learning Integration:** O-RAN deployments have begun integrating AI and machine learning techniques for optimizing network performance, predictive maintenance, and automated troubleshooting. In 6G networks, AI and ML could play an even more significant role in managing complex network resources, mitigating security threats, and enabling intelligent services and applications.
4) **Slicing and Network Customization:** O-RAN's exploration of network slicing to provide differentiated services aligns with the evolving requirements of various 6G use cases, such as eMBB, mMTC, and URLLC. In 6G, network slicing could evolve to support dynamic, on-demand creation of customized slices tailored to specific applications and user needs.
5) **Security and Privacy Considerations:** O-RAN deployments have raised concerns about security and privacy, particularly regarding the integration of components from multiple vendors and vulnerabilities in open interfaces. Addressing these concerns will be crucial in 6G networking, especially with the proliferation of connected devices, sensitive data, and mission-critical applications.
6) **Standardization and Ecosystem Collaboration:** O-RAN initiatives have underscored the importance of industry collaboration, interoperability testing, and standardization efforts to ensure seamless integration and scalability. In the development of 6G networking, fostering an open ecosystem and engaging stakeholders across academia, industry, and government will be essential for driving innovation and accelerating adoption.

In conclusion, while O-RAN and 6G networking serve different objectives and operate on different timelines, insights from the former can inform the development of the latter. Using principles of openness, flexibility, virtualization, intelligence, and collaboration, stakeholders lay the foundation for resilient, efficient, and transformative 6G networks that will shape the future of communications.

XI. LESSONS LEARNED

The research presented in this paper offers valuable insights into the integration of O-RAN architectures with emerging 6G technologies. Below, we outline the key lessons learned from this study:

Energy Efficiency is Crucial for Scaling O-RAN in 6G: One of the critical challenges for O-RAN in 6G networks is ensuring EE. As discussed in Section V on O-RAN architecture, the disaggregation of network components and the use of virtualized functions can create opportunities for energy savings, but it also introduces new demands on power consumption, particularly in DUs and RUs. The integration of hardware accelerators (such as FPGAs and ASICs) and dynamic resource allocation were identified as key enablers for reducing energy consumption. Future O-RAN implementations must focus on optimizing power usage in both hardware and software layers to ensure sustainability in large-scale 6G deployments.

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Table XX
CONCEPTUAL EVALUATIONS OF EMERGING RESEARCH DIRECTIONS

Research Direction	Advantages	Disadvantages
Blockchain for Security	Immutable record, enhanced trust	Latency and scalability issues
Interoperability	Ensures seamless connectivity across legacy and emerging systems	Complexity in achieving compatibility
Energy Efficiency	Reduces costs and carbon footprint	Balancing power-saving modes with network performance
AI for Network Optimization	Predictive analytics, dynamic resource allocation	Model bias, difficulty in integrating with existing network infrastructure

Low Latency Requires Edge Computing and Virtualization: One of the core requirements for 6G is ultra-low latency (ULL), particularly for mission-critical applications such as remote surgery and industrial automation. O-RAN's architecture, with the separation of the control and user planes and the ability to virtualize RAN functions, enables operators to deploy key processing functions closer to the network edge. This allows for faster data processing and reduces the round-trip time for data transmission, as highlighted in Section VI on latency management. The lessons from deploying 5G networks show that without edge computing integration, O-RAN will struggle to meet 6G's stringent latency requirements.

Security Must Be Addressed Holistically in O-RAN: The open and disaggregated nature of O-RAN introduces significant security challenges, as discussed in Section VII. With multiple vendors involved in building RAN components and open interfaces between them, the attack surface for potential cyber threats grows. As highlighted in the survey, AI-driven threat detection and real-time anomaly detection can provide critical security layers, but these solutions must be integrated at multiple levels—from hardware to software. A key lesson learned is that a more comprehensive and proactive security model, which includes quantum-safe encryption techniques for long-term protection, is essential for future O-RAN deployments.

Digital Twin Technology Enhances Network Optimization: The inclusion of DT, as discussed in Section V, offers powerful capabilities for real-time network optimization and performance monitoring. By creating a virtual representation of the network, operators can simulate changes, predict failures, and optimize network parameters without disrupting live operations. DTs have been identified as a crucial tool for enhancing network automation and improving resource allocation in complex 6G environments. However, for this technology to deliver its full potential, it must be integrated seamlessly with O-RAN's modular architecture, which requires additional research on interoperability and scalability.

Massive MIMO Technology Is Key to High Throughput: As O-RAN moves into the 6G era mMIMO will be a cornerstone technology for achieving the required levels of throughput and spectral efficiency. Section VII of the survey highlights that O-RAN's flexibility enables easier integration of mMIMO by decoupling the hardware and software layers, allowing operators to deploy best-in-class solutions from multiple vendors. However, a key lesson learned is that to fully realize the benefits of mMIMO, the fronthaul network connecting the distributed units and radio units must be optimized for handling the massive data flows generated by these systems.

V2X Requires a Modular Approach: V2X communication is a critical use case for 5G and 6G networks and yet it is primarily categorized as a technology. As discussed in Section VII, V2X relies on low latency and high reliability for safety-critical applications in smart transportation. O-RAN's modular architecture offers the flexibility needed to accommodate V2X requirements through dynamic resource allocation and edge computing. The lesson here is that customization of network slices for V2X use cases is essential to ensure seamless and reliable communication in 6G networks.

Challenges Experienced During the Research Process

While conducting this research, the following challenges were encountered:

Data Availability: A lack of publicly available performance data for O-RAN implementations hindered precise evaluations of proposed solutions, especially for 6G-specific scenarios.

Fragmented Literature: The fragmented nature of existing surveys and research papers made it challenging to synthesize a cohesive taxonomy and identify gaps. Integrating insights from disparate sources required substantial effort.

Model Complexity: Simulating real-world O-RAN deployments, particularly for energy efficiency and latency-sensitive applications, required complex models that were difficult to validate without extensive testing environments.

Standardization Conflicts: Reconciling differing terminologies and specifications across standardization bodies (e.g., O-RAN Alliance and 3GPP) posed difficulties in aligning findings and proposing universally applicable solutions.

Emerging Technologies: Incorporating nascent technologies such as quantum computing and dApps presented difficulties due to limited existing research and technical maturity in these areas.

The lessons learned from this study underscore the transformative potential of O-RAN in shaping the future of 6G networks. However, they also reveal the critical need for addressing technical, operational, and standardization challenges to fully realize this potential. Moving forward, these insights provide a roadmap for researchers and practitioners aiming to advance O-RAN for 6G deployments.

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